ASD-STE100 TECHNICAL MATHEMATICS MANUAL

Linear Algebra: Vectors, Matrices, and the Four Subspaces

Complete Equation Derivations from +2 Mathematics, SVG Diagrams, Verbatim Examples, and Full Problem Set Solutions

ASD-STE100 Writing Standard Applied:

- Every instruction uses active voice and imperative verbs (*Add*, *Multiply*, *Substitute*, *Eliminate*).
- Every mathematical equation in the syllabus (Strang Ch. 1.1, 1.3, Ch. 3.1–3.6) is derived step-by-step from Plus-Two (+2) coordinate geometry, simultaneous equations, and calculus.
- All in-text examples, Worked Examples, and Problem Set exercises are copied verbatim with full solutions placed at the end of the page/section.

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Plus-Two (+2) Mathematical Toolkit

This chapter lists the six foundational facts from Plus-Two (+2) high-school mathematics. Every equation in this book is derived directly from these six tools.

T1 Ordered Pairs and Cartesian Coordinates in 2D and 3D

A point P in the xy plane has two real coordinates (x, y). A point Q in xyz space has three real coordinates (x, y, z). Two points are equal if and only if their corresponding coordinates are equal:

$$(x_1, y_1) = (x_2, y_2) \iff x_1 = x_2 \text{ and } y_1 = y_2. \tag{0.1}$$

T2 Displacement Arrows and the Parallelogram Rule

An arrow from the origin $(0, 0)$ to (v_1, v_2) represents a displacement of v_1 units along the x axis and v_2 units along the y axis. If a second displacement (w_1, w_2) starts at the head of (v_1, v_2), the total horizontal displacement is $v_1 + w_1$ and the total vertical displacement is $v_2 + w_2$.

T3 Dot Product and Perpendicularity (Orthogonality)

For two 3D vectors $\boldsymbol{u} = (u_1, u_2, u_3)$ and $\boldsymbol{v} = (v_1, v_2, v_3)$, the dot product is:

$$\boldsymbol{u} \cdot \boldsymbol{v} = u_1v_1 + u_2v_2 + u_3v_3 = \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| \cos\theta. \tag{0.2}$$

Two nonzero vectors are perpendicular ($\theta = 90^\circ$, $\cos 90^\circ = 0$) if and only if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$.

T4 Simultaneous Linear Equations and Elimination

Adding a scalar multiple of one equation to another equation does not change the solution set. If $E_1 = 0$ and $E_2 = 0$, then $E_1 = 0$ and $E_2 - \ell E_1 = 0$ have the exact same solutions.

T5 2x2 Determinants and Invertibility

The system $ax + by = b_1$, $cx + dy = b_2$ has a unique solution for every (b_1, b_2) if and only if the determinant $ad - bc \neq 0$. If $ad - bc = 0$, the two rows are proportional.

T6 Elementary Calculus Derivatives and Telescoping Sums

The derivative of t^n is $\frac{\mathrm{d}}{\mathrm{d}t}(t^n) = nt^{n-1}$. The binomial expansion is $(t+1)^n = t^n + nt^{n-1} + \binom{n}{2}t^{n-2} + \cdots + 1$. A telescoping sum cancels interior terms:

$$(x_1 - 0) + (x_2 - x_1) + (x_3 - x_2) + \cdots + (x_n - x_{n-1}) = x_n. \tag{0.3}$$

MODULE 1

Vectors, Linear Combinations, and Matrices

CHAPTER 1

Vectors and Linear Combinations (Sections 1.1 & 1.3)

Syllabus Coverage: Column vectors in $\mathbf{R}^2$, $\mathbf{R}^3$, and $\mathbf{R}^n$; vector addition $v + w$; scalar multiplication cv; linear combinations $cu + dv + ew$; geometry of lines, planes, and $\mathbf{R}^3$; matrix-vector multiplication Sx; difference matrices A and sum matrices $S = A^{-1}$.

1.1 Vector Addition, Scalar Multiplication, and Linear Combinations

GOAL

Derive the algebraic rules for vector addition $v + w$, scalar multiplication cv, and linear combinations $cv + dw$ from +2 coordinate displacements (T1, T2).

FROM YOUR +2 BASE

From T1 and T2, a point (v_1, v_2) in the plane is reached by moving v_1 units along the x axis and v_2 units along the y axis. Horizontal and vertical steps are independent ("you cannot add apples and oranges").

Step 1: Define the column vector. Write the two separate numbers v_1 (first component) and v_2 (second component) as a single column vector v:

$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \qquad \text{also written inline as } v = (v_1, v_2). \tag{1.1}$$

Note that $v = (v_1, v_2)$ is a column vector temporarily lying down to save space. The row vector $\begin{bmatrix} v_1 & v_2 \end{bmatrix}$ is the **transpose** v^T.

Step 2: Derive vector addition and scalar multiplication. Place the start of $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ at the end of $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. By T2, add the x-displacements together and add the y-displacements together:

$$v + w = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix} \qquad \textit{add component by component} \tag{1.2}$$

$$cv = c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}, \qquad -v = \begin{bmatrix} -v_1 \\ -v_2 \end{bmatrix} \qquad \textit{multiply each component by scalar } c \tag{1.3}$$

When $c = 0$, $0v = \mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. The **zero vector** $\mathbf{0}$ has components 0 and 0. It is not the scalar number 0.

Step 3: Combine addition and scalar multiplication. Multiply v by c, multiply w by d, and add the results. For three vectors u, v, w in $\mathbf{R}^3$:

$$cu + dv + ew = c\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + d\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + e\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} cu_1 + dv_1 + ew_1 \\ cu_2 + dv_2 + ew_2 \\ cu_3 + dv_3 + ew_3 \end{bmatrix} \tag{1.4}$$

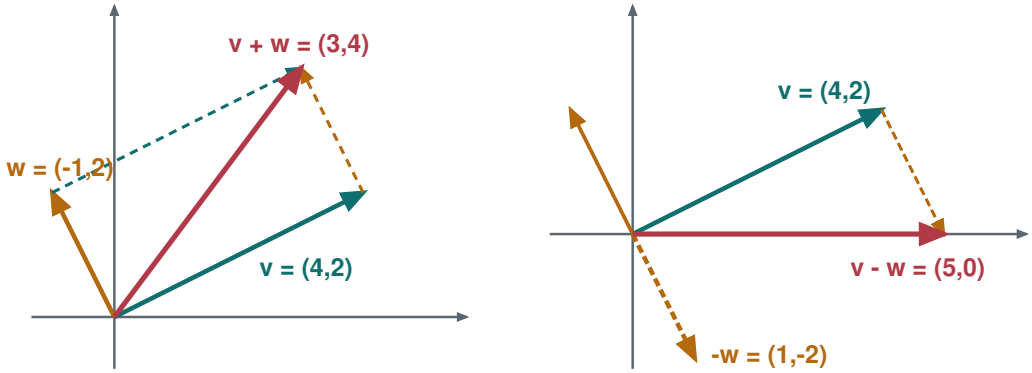


Fig. 1.1 Left: Head-to-tail vector addition $v + w = (3, 4)$ along the parallelogram diagonal. Right: Vector subtraction $v - w = v + (-w) = (5, 0)$.

RESULT (Geometry of All Linear Combinations in $\mathbf{R}^3$)

For typical nonzero vectors u, v, w in $\mathbf{R}^3$:

1. All combinations cu (as c varies over all real numbers) fill a **line** through $(0, 0, 0)$.
2. All combinations $cu + dv$ fill a **plane** through $(0, 0, 0)$ (unless v lies on the line cu).
3. All combinations $cu + dv + ew$ fill **three-dimensional space** $\mathbf{R}^3$ (unless w lies in the plane $cu + dv$).

1.2 Matrix-Vector Multiplication, Sums, and Differences (Section 1.3)

GOAL

Derive matrix-vector multiplication $Sx = b$ from linear combinations of columns, and invert the difference matrix A using telescoping sums (T6).

Step 1: Collect vectors into the columns of a matrix S. Let s_1, s_2, s_3 be three column vectors in $\mathbf{R}^3$, and let $x = (x_1, x_2, x_3)$. Define Sx as the linear combination of the columns of S:

$$Sx = \begin{bmatrix} | & | & | \\ s_1 & s_2 & s_3 \\ | & | & | \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 s_1 + x_2 s_2 + x_3 s_3 = \begin{bmatrix} (\text{row 1 of } S) \cdot x \\ (\text{row 2 of } S) \cdot x \\ (\text{row 3 of } S) \cdot x \end{bmatrix}. \tag{1.5}$$

Step 2: Derive the inverse of the backward difference matrix A. Consider the $n \times n$ backward difference system $A\boldsymbol{x} = \boldsymbol{b}$ (shown here for $n = 4$):

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \\ x_4 - x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}. \tag{1.6}$$

By **T6**, add the first k equations so the interior x_i terms telescope:

$$x_1 = b_1, \quad x_2 = b_1 + b_2, \quad x_3 = b_1 + b_2 + b_3, \quad x_4 = b_1 + b_2 + b_3 + b_4$$

$$\implies \boldsymbol{x} = A^{-1}\boldsymbol{b} = S\boldsymbol{b} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}. \tag{1.7}$$

Thus, the inverse of the lower-triangular **difference matrix** A is the lower-triangular **sum matrix** $S = A^{-1}$.

1.3 In-Text and Worked Examples (Section 1.1)

BOOK EXAMPLE 1.1 In-Text (p. 1–5) Commutativity and 3D Linear Combinations

(1) Check $v+w=w+v$ for $v=\begin{bmatrix}1\\5\end{bmatrix}$ and $w=\begin{bmatrix}3\\3\end{bmatrix}$.

(2) Compute $v+w$ and $v-w$ for $v=\begin{bmatrix}4\\2\end{bmatrix}$ and $w=\begin{bmatrix}-1\\2\end{bmatrix}$.

(3) Compute $v+w$ for $v=\begin{bmatrix}1\\1\\-1\end{bmatrix}$ and $w=\begin{bmatrix}2\\3\\4\end{bmatrix}$.

(4) Question (p. 5): Do the four sides of the parallelogram v, w all lie in the same plane? What vector does $v+w-v-w$ produce? Compute $u+4v-2w$ for $u=(1,0,3)$, $v=(1,2,1)$, $w=(2,3,-1)$.

BOOK EXAMPLE 1.1 A Describing a Plane of Linear Combinations

The linear combinations of $v=(1,1,0)$ and $w=(0,1,1)$ fill a plane. *Describe that plane.* Find a vector that is *not* a combination of v and w.

BOOK EXAMPLE 1.1 B Subsets of Linear Combinations (Lines, Half-Lines, Half-Planes)

For $v=(1,0)$ and $w=(0,1)$, describe all points cv with **(1)** *whole numbers* c **(2)** *nonnegative* $c\geq 0$. Then add all vectors dw and describe all $cv+dw$.

BOOK EXAMPLE 1.1 C Expressing a Vector Equation as Scalar Equations

Find two equations for the unknowns c and d so that the linear combination $cv+dw$ equals the vector b:

$$v=\begin{bmatrix}2\\-1\end{bmatrix} \qquad w=\begin{bmatrix}-1\\2\end{bmatrix} \qquad b=\begin{bmatrix}1\\0\end{bmatrix}.$$

END-OF-PAGE SOLUTION Section 1.1 In-Text & Worked Examples 1.1 A, 1.1 B, 1.1 C

In-Text (1): $v+w=\begin{bmatrix}1+3\\5+3\end{bmatrix}=\begin{bmatrix}4\\8\end{bmatrix}$ and $w+v=\begin{bmatrix}3+1\\3+5\end{bmatrix}=\begin{bmatrix}4\\8\end{bmatrix}$. **(2):** $v+w=\begin{bmatrix}4-1\\2+2\end{bmatrix}=\begin{bmatrix}3\\4\end{bmatrix}$; $v-w=\begin{bmatrix}4-(-1)\\2-2\end{bmatrix}=\begin{bmatrix}5\\0\end{bmatrix}$. **(3):** $v+w=(1+2,1+3,-1+4)=(3,4,3)$. **(4):** Yes, all four sides lie in the plane spanned by v and w. The closed loop $v+w-v-w$ produces the **zero** vector $\mathbf{0}$. Next, $u+4v-2w=\begin{bmatrix}1+4-4\\0+8-6\\3+4+2\end{bmatrix}=\begin{bmatrix}1\\2\\9\end{bmatrix}$.

1.1 A: Write the general combination:

$$cv+dw=c\begin{bmatrix}1\\1\\0\end{bmatrix}+d\begin{bmatrix}0\\1\\1\end{bmatrix}=\begin{bmatrix}c\\c+d\\d\end{bmatrix}.$$

In this plane, the second component $b_2=c+d$ always equals the sum of the first component $b_1=c$ and the third component $b_3=d$ ($b_2=b_1+b_3$, or $n\cdot b=0$ with normal $n=(1,-1,1)$).

Four vectors in the plane are $(0,0,0)$, $(2,3,1)$, $(5,7,2)$, and $(\pi, 2\pi, \pi)$. The vector $(1,2,3)$ is *not* in the plane because $2 \neq 1+3$.

1.1 B: (1) $c\boldsymbol{v} = (c,0)$ for integer c gives equally spaced points $(-2,0),(-1,0),(0,0),(1,0),(2,0)$ along the x axis. **(2)** $c\boldsymbol{v} = (c,0)$ for $c \geq 0$ fills the nonnegative half-line (positive x axis starting at $(0,0)$). **(1')** Adding all $d\boldsymbol{w} = (0,d)$ for real d puts a vertical line through each integer point $(c,0)$, creating infinitely many parallel vertical lines. **(2')** Adding all $d\boldsymbol{w} = (0,d)$ to $c \geq 0$ fills the right half-plane $\{(x,y) : x \geq 0,\ -\infty < y < \infty\}$.

1.1 C: Write the vector equation $c\begin{bmatrix} 2 \\ -1 \end{bmatrix} + d\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Separate into two scalar equations:

$$2c - d = 1 \quad \text{and} \quad -c + 2d = 0 \implies c = 2d \implies 4d - d = 1 \implies d = \frac{1}{3},\ c = \frac{2}{3}.$$

1.4 Problem Set 1.1 (Verbatim Questions and Solutions)

Problems 1–9 are about addition of vectors and linear combinations.

1. Describe geometrically (line, plane, or all of $\mathbf{R}^3$) all linear combinations of

$$\text{(a)}\ \begin{bmatrix}1\\2\\3\end{bmatrix}\ \text{and}\ \begin{bmatrix}3\\6\\9\end{bmatrix} \qquad \text{(b)}\ \begin{bmatrix}1\\0\\0\end{bmatrix}\ \text{and}\ \begin{bmatrix}0\\2\\3\end{bmatrix} \qquad \text{(c)}\ \begin{bmatrix}2\\0\\0\end{bmatrix}\ \text{and}\ \begin{bmatrix}0\\2\\2\end{bmatrix}\ \text{and}\ \begin{bmatrix}2\\2\\3\end{bmatrix}$$

2. Draw $v = \begin{bmatrix}4\\1\end{bmatrix}$ and $w = \begin{bmatrix}-2\\2\end{bmatrix}$ and $v+w$ and $v-w$ in a single xy plane.

3. If $v+w = \begin{bmatrix}5\\1\end{bmatrix}$ and $v-w = \begin{bmatrix}1\\5\end{bmatrix}$, compute and draw v and w.

4. From $v = \begin{bmatrix}2\\1\end{bmatrix}$ and $w = \begin{bmatrix}1\\2\end{bmatrix}$, find the components of $3v+w$ and $cv+dw$.

5. Compute $u+v+w$ and $2u+2v+w$. How do you know u, v, w lie in a plane?

$$\textbf{In a plane} \qquad u = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad v = \begin{bmatrix}-3\\1\\-2\end{bmatrix}, \quad w = \begin{bmatrix}2\\-3\\-1\end{bmatrix}.$$

6. Every combination of $v = (1,-2,1)$ and $w = (0,1,-1)$ has components that add to _____. Find c and d so that $cv+dw = (3,3,-6)$.

7. In the xy plane mark all nine of these linear combinations:

$$c\begin{bmatrix}2\\1\end{bmatrix} + d\begin{bmatrix}0\\1\end{bmatrix} \quad \text{with} \quad c = 0,1,2 \text{ and } d = 0,1,2.$$

8. The parallelogram in Figure 1.1 has diagonal $v+w$. What is its other diagonal? What is the sum of the two diagonals? Draw that vector sum.

9. If three corners of a parallelogram are $(1,1)$, $(4,2)$, and $(1,3)$, what are all three of the possible fourth corners? Draw two of them.

END-OF-PAGE SOLUTION Problem Set 1.1 (Problems 1–9)

1. (a) Line in $\mathbf{R}^3$ because $(3,6,9) = 3(1,2,3)$. **(b) Plane** in $\mathbf{R}^3$ because the two vectors are not parallel. **(c) All of $\mathbf{R}^3$** because $(2,2,3)$ is not a combination of $(2,0,0)$ and $(0,2,2)$ (combinations of the first two have $b_2 = b_3$, whereas $2 \neq 3$).

2. $v+w = \begin{bmatrix}4-2\\1+2\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$ and $v-w = \begin{bmatrix}4-(-2)\\1-2\end{bmatrix} = \begin{bmatrix}6\\-1\end{bmatrix}$. Plot arrows from $(0,0)$ to $(4,1)$, $(-2,2)$, $(2,3)$, and $(6,-1)$.

3. Add the two equations: $2v = \begin{bmatrix}6\\6\end{bmatrix} \implies v = \begin{bmatrix}3\\3\end{bmatrix}$. Subtract the two equations: $2w = \begin{bmatrix}4\\-4\end{bmatrix} \implies w = \begin{bmatrix}2\\-2\end{bmatrix}$.

4. $3v+w = \begin{bmatrix}6+1\\3+2\end{bmatrix} = \begin{bmatrix}7\\5\end{bmatrix}$; $cv+dw = \begin{bmatrix}2c+d\\c+2d\end{bmatrix}$.

5. $\boldsymbol{u}+\boldsymbol{v}+\boldsymbol{w}=\begin{bmatrix}1-3+2\\2+1-3\\3-2-1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$. Then $2\boldsymbol{u}+2\boldsymbol{v}+\boldsymbol{w}=2(\boldsymbol{u}+\boldsymbol{v}+\boldsymbol{w})-\boldsymbol{w}=-\boldsymbol{w}=\begin{bmatrix}-2\\3\\1\end{bmatrix}$. Because $\boldsymbol{w}=-\boldsymbol{u}-\boldsymbol{v}$ is a linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$, all three vectors lie in the plane spanned by $\boldsymbol{u}$ and $\boldsymbol{v}$.

6. The components of $\boldsymbol{v}$ add to $1-2+1=0$, and the components of $\boldsymbol{w}$ add to $0+1-1=0$. Thus every combination $c\boldsymbol{v}+d\boldsymbol{w}=(c,-2c+d,c-d)$ has components that add to $\mathbf{0}$. From $c=3$ and $-2c+d=3 \implies -6+d=3 \implies d=9$. Check third component: $c-d=3-9=-6$. So $c=3, d=9$.

7. For $c\in\{0,1,2\}$ and $d\in\{0,1,2\}$, the 9 points are: $c=0$: $(0,0),(0,1),(0,2)$; $c=1$: $(2,1),(2,2),(2,3)$; $c=2$: $(4,2),(4,3),(4,4)$.

8. The other diagonal is $\boldsymbol{v}-\boldsymbol{w}$ (or $\boldsymbol{w}-\boldsymbol{v}$). The sum of the two diagonals is $(\boldsymbol{v}+\boldsymbol{w})+(\boldsymbol{v}-\boldsymbol{w})=2\boldsymbol{v}$ (or $(\boldsymbol{v}+\boldsymbol{w})+(\boldsymbol{w}-\boldsymbol{v})=2\boldsymbol{w}$).

9. Let $A=(1,1)$, $B=(4,2)$, $C=(1,3)$. The three possible fourth corners are $A+B-C=(4,0)$, $A+C-B=(-2,2)$, and $B+C-A=(4,4)$.

Problems 10–14 are about special vectors on cubes and clocks in Figure 1.4.

10. Which point of the cube is $\boldsymbol{i}+\boldsymbol{j}$? Which point is the vector sum of $\boldsymbol{i}=(1,0,0)$ and $\boldsymbol{j}=(0,1,0)$ and $\boldsymbol{k}=(0,0,1)$? Describe all points (x,y,z) in the cube.

11. Four corners of the cube are $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$. What are the other four corners? Find the coordinates of the center point of the cube. The center points of the six faces are _____.

12. How many corners does a cube have in 4 dimensions? How many 3D faces? How many edges? A typical corner is $(0,0,1,0)$. A typical edge goes to $(0,1,0,0)$ [correction: $(0,1,1,0)$].

13. **(a)** What is the sum $\boldsymbol{V}$ of the twelve vectors that go from the center of a clock to the hours 1:00, 2:00, ..., 12:00?
(b) If the 2:00 vector is removed, why do the 11 remaining vectors add to 8:00?
(c) What are the components of that 2:00 vector $\boldsymbol{v}=(\cos\theta,\sin\theta)$?

14. Suppose the twelve vectors start from 6:00 at the bottom instead of $(0,0)$ at the center. The vector to 12:00 is doubled to $(0,2)$. Add the new twelve vectors.

Problems 15–19 go further with linear combinations of v and w (Figure 1.5a).

15. Figure 1.5a shows $\frac{1}{2}\boldsymbol{v}+\frac{1}{2}\boldsymbol{w}$. Mark the points $\frac{3}{4}\boldsymbol{v}+\frac{1}{4}\boldsymbol{w}$ and $\frac{1}{4}\boldsymbol{v}+\frac{1}{4}\boldsymbol{w}$ and $\boldsymbol{v}+\boldsymbol{w}$.

16. Mark the point $-\boldsymbol{v}+2\boldsymbol{w}$ and any other combination $c\boldsymbol{v}+d\boldsymbol{w}$ with $c+d=1$. Draw the line of all combinations that have $c+d=1$.

17. Locate $\frac{1}{3}\boldsymbol{v}+\frac{1}{3}\boldsymbol{w}$ and $\frac{2}{3}\boldsymbol{v}+\frac{2}{3}\boldsymbol{w}$. The combinations $c\boldsymbol{v}+c\boldsymbol{w}$ fill out what line?

18. Restricted by $0\le c\le 1$ and $0\le d\le 1$, shade in all combinations $c\boldsymbol{v}+d\boldsymbol{w}$.

19. Restricted only by $c\ge 0$ and $d\ge 0$ draw the "cone" of all combinations $c\boldsymbol{v}+d\boldsymbol{w}$.

END-OF-PAGE SOLUTION Problem Set 1.1 (Problems 10–19)

10. $\boldsymbol{i}+\boldsymbol{j}=(1,1,0)$ (bottom corner opposite $(0,0,0)$ in the xy face). $\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k}=(1,1,1)$ (corner opposite $(0,0,0)$ across the cube). All points in the solid unit cube satisfy $0\le x\le 1$, $0\le y\le 1$, $0\le z\le 1$.

11. The other four corners are $(1,1,0)$, $(1,0,1)$, $(0,1,1)$, and $(1,1,1)$. The cube center is $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$. The six face centers have one coordinate equal to 0 or 1 and the other two equal to $\frac{1}{2}$: $(0,\frac{1}{2},\frac{1}{2})$, $(1,\frac{1}{2},\frac{1}{2})$, $(\frac{1}{2},0,\frac{1}{2})$, $(\frac{1}{2},1,\frac{1}{2})$, $(\frac{1}{2},\frac{1}{2},0)$, $(\frac{1}{2},\frac{1}{2},1)$.

12. A 4D hypercube has $2^4=\mathbf{16}$ corners (all binary 4-tuples), $2\times 4=\mathbf{8}$ three-dimensional faces (fix one coordinate to 0 or 1), and $\frac{16\times 4}{2}=\mathbf{32}$ edges (each of the 16 corners connects to 4 neighbors differing in one coordinate).

13. **(a)** $\boldsymbol{V}=\mathbf{0}=(0,0)$ because each hour vector cancels its opposite hour vector (1:00 cancels 7:00, etc.). **(b)** Removing $\boldsymbol{v}_{2:00}$ leaves $\boldsymbol{V}-\boldsymbol{v}_{2:00}=-\boldsymbol{v}_{2:00}=\boldsymbol{v}_{8:00}$, which points to 8:00. **(c)** 2:00 is at angle $\theta=30^\circ=\frac{\pi}{6}$ above the positive x axis (3:00), so $\boldsymbol{v}=(\cos 30^\circ,\sin 30^\circ)=(\frac{\sqrt{3}}{2},\frac{1}{2})$.

14. Shifting the origin from the center $(0,0)$ to 6:00 $(0,-1)$ adds the vector $(0,1)$ to each of the 12 original clock vectors $\boldsymbol{v}_k$. Therefore the new sum is $\sum_{k=1}^{12}(\boldsymbol{v}_k+(0,1))=\mathbf{0}+12(0,1)=(0,12)$.

15. $\frac{3}{4}\boldsymbol{v}+\frac{1}{4}\boldsymbol{w}$ is three-quarters of the way along the dashed segment from $\boldsymbol{w}$ to $\boldsymbol{v}$; $\frac{1}{4}\boldsymbol{v}+\frac{1}{4}\boldsymbol{w}$ is halfway from $\mathbf{0}$ to $\boldsymbol{u}=\frac{1}{2}\boldsymbol{v}+\frac{1}{2}\boldsymbol{w}$; $\boldsymbol{v}+\boldsymbol{w}$ is the opposite corner of the parallelogram ($2\boldsymbol{u}$).

16. $-\boldsymbol{v}+2\boldsymbol{w}=\boldsymbol{w}+(\boldsymbol{w}-\boldsymbol{v})$ lies on the line through the tips of $\boldsymbol{v}$ and $\boldsymbol{w}$, extended past $\boldsymbol{w}$ by one segment length. All combinations with $c+d=1$ form the infinite line passing through the tips of $\boldsymbol{v}$ and $\boldsymbol{w}$.

17. $\frac{1}{3}\boldsymbol{v}+\frac{1}{3}\boldsymbol{w}$ and $\frac{2}{3}\boldsymbol{v}+\frac{2}{3}\boldsymbol{w}$ lie at $\frac{1}{3}$ and $\frac{2}{3}$ along the main diagonal from $\mathbf{0}$ to $\boldsymbol{v}+\boldsymbol{w}$. All $c\boldsymbol{v}+c\boldsymbol{w}=c(\boldsymbol{v}+\boldsymbol{w})$ fill the infinite line through $\mathbf{0}$ and $\boldsymbol{u}=\frac{1}{2}\boldsymbol{v}+\frac{1}{2}\boldsymbol{w}$.

18. $0\le c\le 1$ and $0\le d\le 1$ fills the entire parallelogram with vertices $\mathbf{0},\boldsymbol{v},\boldsymbol{w},\boldsymbol{v}+\boldsymbol{w}$.

19. $c \geq 0$ and $d \geq 0$ fills the infinite sector (2D cone) bounded by the rays $c\boldsymbol{v}$ ($c \geq 0$) and $d\boldsymbol{w}$ ($d \geq 0$).

Problems 20–25 deal with u, v, w in three-dimensional space (see Figure 1.5b).

20. Locate $\frac{1}{3}u + \frac{1}{3}v + \frac{1}{3}w$ and $\frac{1}{2}u + \frac{1}{2}w$ in Figure 1.5b. Challenge problem: Under what restrictions on c, d, e, will the combinations $cu + dv + ew$ fill in the dashed triangle? To stay in the triangle, one requirement is $c \geq 0, d \geq 0, e \geq 0$.

21. The three sides of the dashed triangle are $v - u$ and $w - v$ and $u - w$. Their sum is _____. Draw the head-to-tail addition around a plane triangle of $(3,1)$ plus $(-1,1)$ plus $(-2,-2)$.

22. Shade in the pyramid of combinations $cu + dv + ew$ with $c \geq 0, d \geq 0, e \geq 0$ and $c + d + e \leq 1$. Mark the vector $\frac{1}{2}(u + v + w)$ as inside or outside this pyramid.

23. If you look at *all* combinations of those u, v, and w, is there any vector that can't be produced from $cu + dv + ew$? Different answer if u, v, w are all in _____.

24. Which vectors are combinations of u and v, and also combinations of v and w?

25. Draw vectors u, v, w so that their combinations $cu + dv + ew$ fill only a line. Find vectors u, v, w so that their combinations $cu + dv + ew$ fill only a plane.

26. What combination $c \begin{bmatrix} 1 \\ 2 \end{bmatrix} + d \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ produces $\begin{bmatrix} 14 \\ 8 \end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.

27. *Review Question.* In xyz space, where is the plane of all linear combinations of $i = (1,0,0)$ and $i + j = (1,1,0)$?

28. **Challenge Problems:** Find vectors v and w so that $v + w = (4,5,6)$ and $v - w = (2,5,8)$. This is a question with _____ unknown numbers, and an equal number of equations to find those numbers.

29. Find two different combinations of the three vectors $u = (1,3)$ and $v = (2,7)$ and $w = (1,5)$ that produce $b = (0,1)$. Slightly delicate question: If I take *any* three vectors u, v, w in the plane, will there always be two different combinations that produce $b = (0,1)$?

30. The linear combinations of $v = (a,b)$ and $w = (c,d)$ fill the plane unless _____. Find four vectors u, v, w, z with four components each so that their combinations $cu + dv + ew + fz$ produce all vectors (b_1, b_2, b_3, b_4) in four-dimensional space.

31. Write down three equations for c, d, e so that $cu + dv + ew = b$. Can you somehow find c, d, and e?

$$u = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \qquad v = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} \qquad w = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \qquad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

END-OF-PAGE SOLUTION Problem Set 1.1 (Problems 20–31)

20. $\frac{1}{3}u + \frac{1}{3}v + \frac{1}{3}w$ is the centroid of the dashed triangle; $\frac{1}{2}u + \frac{1}{2}w$ is the midpoint of side uw. The dashed triangle is filled when $c \geq 0, d \geq 0, e \geq 0$ and $\mathbf{c + d + e = 1}$.

21. $(v - u) + (w - v) + (u - w) = \mathbf{0}$. Similarly $(3,1) + (-1,1) + (-2,-2) = (0,0)$ forms a closed triangle.

22. The pyramid has vertices $\mathbf{0}, u, v, w$. For $\frac{1}{2}(u + v + w)$, $c + d + e = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$, so it is **outside** the pyramid.

23. For typical non-coplanar u, v, w, every vector in $\mathbf{R}^3$ can be produced. Different answer if u, v, w are all in **the same plane** (or on the same line).

24. The intersection of the uv plane and the vw plane is the line of all scalar multiples dv.

25. Line: choose $v = 2u$ and $w = 3u$ along one line. Plane: choose $u = (1,0,0)$, $v = (0,1,0)$, and $w = (1,1,0) = u + v$.

26. Equations: $c + 3d = 14$ and $2c + d = 8$. Subtract $2 \times$ (eq 1) from (eq 2): $-5d = -20 \implies d = 4,\ c = 2$.

27. Because $c(1,0,0)+d(1,1,0)=(c+d,d,0)$, the plane is the **$xy$ plane** ($z=0$).
28. Add and divide by 2: $\boldsymbol{v}=(3,5,7)$. Subtract and divide by 2: $\boldsymbol{w}=(1,0,-1)$. There are **6** unknown numbers.
29. Combination 1 (set $e=0$): $-7\boldsymbol{u}+3\boldsymbol{v}+0\boldsymbol{w}=(-7+6,-21+21)=(0,1)$? Wait: $c+2d=0, 3c+7d=1 \implies c=-2, d=1$, so $-2\boldsymbol{u}+\boldsymbol{v}+0\boldsymbol{w}=(0,1)$. Combination 2 (set $d=0$): $c+e=0, 3c+5e=1 \implies c=-\frac{1}{2}, e=\frac{1}{2}$, so $-\frac{1}{2}\boldsymbol{u}+0\boldsymbol{v}+\frac{1}{2}\boldsymbol{w}=(0,1)$. Delicate question: **No**—if $\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}$ all lie on the same line not containing $(0,1)$ (e.g., along $(1,0)$), no combination can produce $(0,1)$.
30. Unless $\boldsymbol{v}$ and $\boldsymbol{w}$ lie on the same line through $(0,0)$ ($ad-bc=0$). In $\mathbf{R}^4$, take the standard basis $\boldsymbol{u}=(1,0,0,0)$, $\boldsymbol{v}=(0,1,0,0)$, $\boldsymbol{w}=(0,0,1,0)$, $\boldsymbol{z}=(0,0,0,1)$.
31. Equations: $2c-d=1$, $-c+2d-e=0$, $-d+2e=0$. From eq 3, $d=2e$. From eq 1, $2c=1+2e$. Substitute into eq 2 multiplied by 2: $-(1+2e)+8e-2e=0 \implies 4e=1 \implies e=\frac{1}{4},\ d=\frac{1}{2},\ c=\frac{3}{4}$.

1.5 Problem Set 1.3 (Verbatim Questions and Solutions)

1. Find the linear combination $2\boldsymbol{s}_1 + 3\boldsymbol{s}_2 + 4\boldsymbol{s}_3 = \boldsymbol{b}$. Then write $\boldsymbol{b}$ as a matrix-vector multiplication $S\boldsymbol{x}$. Compute the dot products $(\text{row of } S) \cdot \boldsymbol{x}$:

$$\boldsymbol{s}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \boldsymbol{s}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad \boldsymbol{s}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ go into the columns of } S.$$

2. Solve these equations $S\boldsymbol{y} = \boldsymbol{b}$ with $\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3$ in the columns of S:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}.$$

The sum of the first n odd numbers is _____.

3. Solve these three equations for y_1, y_2, y_3 in terms of B_1, B_2, B_3:

$$S\boldsymbol{y} = \boldsymbol{B} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix}.$$

Write the solution $\boldsymbol{y}$ as a matrix $A = S^{-1}$ times the vector $\boldsymbol{B}$. Are the columns of S independent or dependent?

4. Find a combination $x_1\boldsymbol{w}_1 + x_2\boldsymbol{w}_2 + x_3\boldsymbol{w}_3$ that gives the zero vector:

$$\boldsymbol{w}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \qquad \boldsymbol{w}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \qquad \boldsymbol{w}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Those vectors are (independent) (dependent). The three vectors lie in a _____. The matrix W with those columns is *not invertible*.

5. The rows of that matrix W produce three vectors (I write them as columns):

$$\boldsymbol{r}_1 = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} \qquad \boldsymbol{r}_2 = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} \qquad \boldsymbol{r}_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}.$$

Linear algebra says that these vectors must also lie in a plane. There must be many combinations with $y_1\boldsymbol{r}_1 + y_2\boldsymbol{r}_2 + y_3\boldsymbol{r}_3 = \boldsymbol{0}$. Find two sets of y's.

6. Which values of c give dependent columns (combination equals zero)?

$$\begin{bmatrix} 1 & 3 & 5 \\ 1 & 2 & 4 \\ 1 & 1 & c \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & c \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} c & c & c \\ 2 & 1 & 5 \\ 3 & 3 & 6 \end{bmatrix}$$

7. If the columns combine into $A\boldsymbol{x} = \boldsymbol{0}$ then each row has $\boldsymbol{r} \cdot \boldsymbol{x} = 0$:

$$\begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \boldsymbol{a}_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \textbf{By rows} \quad \begin{bmatrix} \boldsymbol{r}_1 \cdot \boldsymbol{x} \\ \boldsymbol{r}_2 \cdot \boldsymbol{x} \\ \boldsymbol{r}_3 \cdot \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to $\boldsymbol{x}$?

END-OF-PAGE SOLUTION Problem Set 1.3 (Problems 1–7)

1. $\boldsymbol{b} = 2\begin{bmatrix}1\\1\\1\end{bmatrix} + 3\begin{bmatrix}0\\1\\1\end{bmatrix} + 4\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}2\\5\\9\end{bmatrix} = \begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}2\\3\\4\end{bmatrix}$. Dot products: $(1,0,0)\cdot(2,3,4) = 2$; $(1,1,0)\cdot(2,3,4) = 5$; $(1,1,1)\cdot(2,3,4) = 9$.

2. First system: $y_1 = 1, y_2 = 0, y_3 = 0$. Second system: $y_1 = 1, y_2 = 4-1 = 3, y_3 = 9-4 = 5$. The sum of the first n odd numbers is $\mathbf{n^2}$.

3. $y_1 = B_1$, $y_2 = B_2 - B_1$, $y_3 = B_3 - B_2$, so $\boldsymbol{y} = A\boldsymbol{B} = \begin{bmatrix}1&0&0\\-1&1&0\\0&-1&1\end{bmatrix}\begin{bmatrix}B_1\\B_2\\B_3\end{bmatrix}$. Columns of S are **independent**.

4. $1\boldsymbol{w}_1 - 2\boldsymbol{w}_2 + 1\boldsymbol{w}_3 = \mathbf{0}$. The vectors are **dependent** and lie in a **plane**.

5. $\boldsymbol{r}_1 - 2\boldsymbol{r}_2 + \boldsymbol{r}_3 = \mathbf{0}$, so $(y_1, y_2, y_3) = (1,-2,1)$ and $(2,-4,2)$ (any nonzero multiple).

6. Matrix 1: row 1 + row 3 = 2 × (row 2) when $5 + c = 2(4) \implies \mathbf{c = 3}$. Matrix 2: col 1 − col 2 = col 3 when $\mathbf{c = -1}$. Matrix 3: row 3 = $\frac{3}{2}$ × (row 2)? No, if $\mathbf{c = 0}$, row 1 is all zeros so columns are dependent.

7. Because $\boldsymbol{r}_1 \cdot \boldsymbol{x} = 0$, $\boldsymbol{r}_2 \cdot \boldsymbol{x} = 0$, and $\boldsymbol{r}_3 \cdot \boldsymbol{x} = 0$, every row vector (and every linear combination in their plane) has zero dot product with $\boldsymbol{x}$ (**T3**).

8. Moving to a 4 by 4 difference equation $A\boldsymbol{x} = \boldsymbol{b}$, find the four components x_1, x_2, x_3, x_4. Then write this solution as $\boldsymbol{x} = S\boldsymbol{b}$ to find the inverse matrix $S = A^{-1}$:

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \boldsymbol{b}.$$

9. What is the *cyclic* 4 by 4 difference matrix C? It will have 1 and -1 in each row. Find all solutions $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ to $C\boldsymbol{x} = \boldsymbol{0}$. The four columns of C lie in a "three-dimensional hyperplane" inside four-dimensional space.

10. A *forward* difference matrix Δ is upper triangular:

$$\Delta\boldsymbol{z} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} z_2 - z_1 \\ z_3 - z_2 \\ 0 - z_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \boldsymbol{b}.$$

Find z_1, z_2, z_3 from b_1, b_2, b_3. What is the inverse matrix in $\boldsymbol{z} = \Delta^{-1}\boldsymbol{b}$?

11. Show that the forward differences $(t+1)^2 - t^2$ are $2t + 1 = \textit{odd numbers}$. As in calculus, the difference $(t+1)^n - t^n$ will begin with the derivative of t^n, which is _____.

12. The last lines of the Worked Example say that the 4 by 4 centered difference matrix in (16) is *invertible*. Solve $C\boldsymbol{x} = (b_1, b_2, b_3, b_4)$ to find its inverse in $\boldsymbol{x} = C^{-1}\boldsymbol{b}$.

13. **Challenge Problems:** The very last words say that the 5 by 5 centered difference matrix is *not invertible*. Write down the 5 equations $C\boldsymbol{x} = \boldsymbol{b}$. Find a combination of left sides that gives zero. What combination of b_1, b_2, b_3, b_4, b_5 must be zero? (The 5 columns lie on a "4-dimensional hyperplane" in 5-dimensional space.)

14. If (a, b) is a multiple of (c, d) with $abcd \neq 0$, show that (a, c) is a multiple of (b, d). This is surprisingly important; two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has dependent columns when it has dependent rows.

END-OF-PAGE SOLUTION Problem Set 1.3 (Problems 8–14)

8. $x_1 = b_1$, $x_2 = b_1 + b_2$, $x_3 = b_1 + b_2 + b_3$, $x_4 = b_1 + b_2 + b_3 + b_4$, and $S = A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$.

9. $C = \begin{bmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$. $C\boldsymbol{x} = \boldsymbol{0}$ requires $x_1 = x_4 = x_3 = x_2$, so all solutions are $\boldsymbol{x} = c(1, 1, 1, 1)$.

10. From bottom to top: $z_3 = -b_3$, $z_2 = z_3 - b_2 = -b_2 - b_3$, $z_1 = z_2 - b_1 = -b_1 - b_2 - b_3$. Thus $\Delta^{-1} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$.

11. $(t+1)^2 - t^2 = (t^2 + 2t + 1) - t^2 = 2t + 1$. By the binomial theorem (T6), $(t+1)^n - t^n$

begins with the derivative of t^n, which is $\mathbf{nt^{n-1}}$.

12. The 4×4 centered difference equations are $x_2 = b_1$, $x_3 - x_1 = b_2$, $x_4 - x_2 = b_3$, $-x_3 = b_4$. Solving gives $x_3 = -b_4$, $x_1 = x_3 - b_2 = -b_2 - b_4$, $x_2 = b_1$, $x_4 = x_2 + b_3 = b_1 + b_3$. Thus:

$$C^{-1} = \begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

13. The 5 equations are $x_2 = b_1$, $x_3 - x_1 = b_2$, $x_4 - x_2 = b_3$, $x_5 - x_3 = b_4$, $-x_4 = b_5$. Add equations 1, 3, and 5: $x_2 + (x_4 - x_2) + (-x_4) = 0$. Thus $\mathbf{b_1 + b_3 + b_5 = 0}$ is required for a solution.

14. If $(a, b) = k(c, d)$, then $a = kc$ and $b = kd$, so $\frac{a}{b} = \frac{kc}{kd} = \frac{c}{d} = m$. Therefore $a = mb$ and $c = md$, which means the first column $(a, c) = m(b, d)$ is a multiple of the second column.

Module 1: Unique Solved Problem Archetypes & Formula Sheet

1. Vector Operations and Coplanarity Test

- **Linear Combination:** $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w} = (cu_1 + dv_1 + ew_1,\ cu_2 + dv_2 + ew_2,\ cu_3 + dv_3 + ew_3)$.
- **Plane Equation from Spanning Vectors (Ex 1.1 A, PS 1.1 #6):** Eliminate c and d from $\boldsymbol{b} = c\boldsymbol{v} + d\boldsymbol{w}$ to obtain the single linear constraint $n_1b_1 + n_2b_2 + n_3b_3 = 0$.
- **Affine Combinations (PS 1.1 #16, #20):** $c\boldsymbol{v} + d\boldsymbol{w}$ with $c + d = 1$ fills the line through the tips of $\boldsymbol{v}$ and $\boldsymbol{w}$. In 3D, $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w}$ with $c + d + e = 1$ and $c, d, e \geq 0$ fills the triangle with vertices $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$.

2. Difference Matrices and Inverse Sum Matrices (PS 1.3 #2, #3, #8, #10, #12)

- **Backward Difference & Sum:** $A\boldsymbol{x} = (x_1, x_2 - x_1, \ldots, x_n - x_{n-1})^T = \boldsymbol{b} \iff x_k = \sum_{i=1}^{k} b_i$.
- **Singular Cyclic / Odd Centered Difference (PS 1.3 #9, #13):** When row combinations sum to $\mathbf{0}^T$, the matrix is not invertible and $\boldsymbol{b}$ must satisfy the same zero-sum condition.

MODULE 2

Vector Spaces, Subspaces, Nullspace, and Rank

CHAPTER 2

Spaces of Vectors, Nullspace, and Rank (Sections 3.1–3.3)

Syllabus Coverage: Vector spaces $\mathbf{R}^n$, $\mathbf{C}^n$, $\mathbf{M}$, $\mathbf{F}$, $\mathbf{P}_n$, $\mathbf{Z}$ and the 8 axioms; subspaces and closure under linear combinations; column space $C(A)$ and span of a set $\mathbf{S}$; nullspace $N(A)$, elimination to U and $R = \text{rref}(A)$, pivot and free variables, special solutions, and nullspace matrix N; rank r, rank-one matrices $A = \boldsymbol{u}\boldsymbol{v}^T$, and pivot columns.

2.1 Spaces of Vectors, Subspaces, and Column Space (Section 3.1)

GOAL

State the eight axioms of a vector space, derive the single combined closure rule for a subspace, and prove that $A\boldsymbol{x} = \boldsymbol{b}$ is solvable if and only if $\boldsymbol{b}$ belongs to the column space $C(A)$.

Step 1: Define a real vector space and the 8 axioms. A real **vector space** $\mathbf{V}$ is a set of vectors with rules for addition $\boldsymbol{x} + \boldsymbol{y}$ and scalar multiplication $c\boldsymbol{x}$ (for real c) that stay inside $\mathbf{V}$ and obey eight axioms:

$$\begin{aligned} (1)\ \boldsymbol{x}+\boldsymbol{y} &= \boldsymbol{y}+\boldsymbol{x} & (5)\ 1\boldsymbol{x} &= \boldsymbol{x} \\ (2)\ \boldsymbol{x}+(\boldsymbol{y}+\boldsymbol{z}) &= (\boldsymbol{x}+\boldsymbol{y})+\boldsymbol{z} & (6)\ (c_1c_2)\boldsymbol{x} &= c_1(c_2\boldsymbol{x}) \\ (3)\ \exists!\ \mathbf{0}:\ \boldsymbol{x}+\mathbf{0} &= \boldsymbol{x} & (7)\ c(\boldsymbol{x}+\boldsymbol{y}) &= c\boldsymbol{x}+c\boldsymbol{y} \\ (4)\ \exists!\ -\boldsymbol{x}:\ \boldsymbol{x}+(-\boldsymbol{x}) &= \mathbf{0} & (8)\ (c_1+c_2)\boldsymbol{x} &= c_1\boldsymbol{x}+c_2\boldsymbol{x}. \end{aligned} \tag{2.1}$$

Important vector spaces include $\mathbf{R}^n$ (real n-vectors), $\mathbf{M}$ (all 2×2 matrices), $\mathbf{F}$ (all real functions $f(x)$), $\mathbf{P}_n$ (polynomials $a_0 + a_1x + \cdots + a_nx^n$ of degree $\le n$), and $\mathbf{Z} = \{\mathbf{0}\}$ (zero vector alone, dimension 0).

Step 2: Derive the Subspace Closure Rule. A **subspace** of a vector space is a nonempty subset (including $\mathbf{0}$) that satisfies two rules for all $\boldsymbol{v}, \boldsymbol{w}$ in the subspace and all scalars c, d:

$$\text{(i) } \boldsymbol{v}+\boldsymbol{w} \text{ is in the subspace, and (ii) } c\boldsymbol{v} \text{ is in the subspace} \iff \boxed{c\boldsymbol{v}+d\boldsymbol{w} \text{ is in the subspace.}} \tag{2.2}$$

Setting $c = 0$ in rule (ii) proves a mandatory fact: **Every subspace contains the zero vector 0.** In $\mathbf{R}^3$, the only possible subspaces are: **(Z)** the single point $(0,0,0)$, **(L)** any line through $(0,0,0)$, **(P)** any plane through $(0,0,0)$, and **($\mathbf{R}^3$)** the whole space.

Step 3: Define the Column Space $C(A)$ and Span. For an $m\times n$ matrix $A = \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix}$, each column has m components (so $\boldsymbol{a}_j \in \mathbf{R}^m$). By (1.5), $A\boldsymbol{x} = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n$. The **column space** $C(A)$ consists of all linear combinations of the columns of A:

$$\boxed{C(A) = \{A\boldsymbol{x} : \boldsymbol{x} \in \mathbf{R}^n\} \subseteq \mathbf{R}^m.} \tag{2.3}$$

Therefore, **the system $Ax = b$ is solvable if and only if b is in the column space $C(A)$.** More generally, for any set $\mathbf{S} = \{v_1, \ldots, v_N\}$ in a vector space $\mathbf{V}$, the subspace of all linear combinations $c_1 v_1 + \cdots + c_N v_N$ is the **span** **SS** of **S** (the smallest subspace containing **S**).

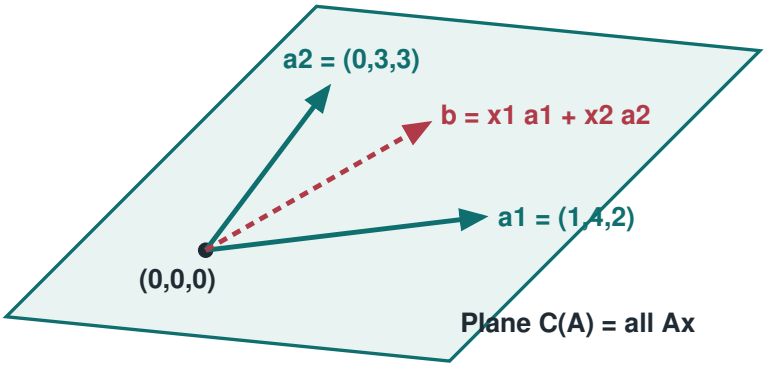


Fig. 2.1 The column space $C(A)$ of a 3×2 matrix A is a plane through $(0,0,0)$ in $\mathbf{R}^3$. $Ax = b$ is solvable exactly when b lies on that plane.

2.2 In-Text and Worked Examples (Section 3.1)

BOOK EXAMPLE 3.1 In-Text 1–5 (p. 123–125) Testing Subspaces and Column Spaces

Ex 1: Keep only vectors $(x, y) \in \mathbf{R}^2$ with $x \geq 0, y \geq 0$ (quarter-plane). Is this a subspace?
Ex 2: Include also vectors with $x \leq 0, y \leq 0$ (two quarter-planes). Is this a subspace?
Ex 3: Inside $\mathbf{M}$ (2×2 matrices), check whether upper triangular matrices $\mathbf{U} = \left\{ \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} \right\}$, diagonal matrices $\mathbf{D} = \left\{ \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \right\}$, and multiples cI are subspaces. Is I alone a subspace?
Ex 4: For $A = \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix}$, describe $C(A)$ and give the always-available solution to $A\boldsymbol{x} = \mathbf{0}$.
Ex 5: Describe the column spaces (subspaces of $\mathbf{R}^2$) for $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$. Find two solutions to $B\boldsymbol{x} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$.

BOOK EXAMPLE 3.1 A Constructing a Matrix with Prescribed Solvability

We are given three different vectors $\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3$. Construct a matrix so that the equations $A\boldsymbol{x} = \boldsymbol{b}_1$ and $A\boldsymbol{x} = \boldsymbol{b}_2$ are solvable, but $A\boldsymbol{x} = \boldsymbol{b}_3$ is *not* solvable. How can you decide if this is possible? How could you construct A?

BOOK EXAMPLE 3.1 B Nested Subspaces and Dual Descriptions

Describe a subspace **S** of each vector space **V**, and then a subspace **SS** of **S**.

$\mathbf{V}_1 =$ all combinations of $(1,1,0,0)$ and $(1,1,1,0)$ and $(1,1,1,1)$
$\mathbf{V}_2 =$ all vectors perpendicular to $\boldsymbol{u} = (1,2,1)$, so $\boldsymbol{u} \cdot \boldsymbol{v} = 0$
$\mathbf{V}_3 =$ all symmetric 2 by 2 matrices (a subspace of **M**)
$\mathbf{V}_4 =$ all solutions to the equation $\mathrm{d}^4y/\mathrm{d}x^4 = 0$ (a subspace of **F**)

Describe each **V** two ways: *All combinations of . . . , all solutions of the equations . . .*

END-OF-PAGE SOLUTION Section 3.1 In-Text 1–5 & Worked Examples 3.1 A, 3.1 B

Ex 1: No. $(2,3)$ is in the quarter-plane, but $(-1)(2,3) = (-2,-3)$ is not (rule (ii) fails).
Ex 2: No. Rule (ii) holds, but $\boldsymbol{v} = (2,3)$ and $\boldsymbol{w} = (-3,-2)$ add to $\boldsymbol{v} + \boldsymbol{w} = (-1,1)$, which is outside both quarter-planes (rule (i) fails).
Ex 3: **U**, **D**, and $\{cI\}$ are all closed under addition and scalar multiplication, so they are subspaces ($\{cI\} \subset \mathbf{D} \subset \mathbf{U} \subset \mathbf{M}$). The single matrix I is **not** a subspace because $0I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \neq I$.
Ex 4: $C(A)$ is the plane in $\mathbf{R}^3$ containing all $x_1(1,4,2) + x_2(0,3,3)$. The always-available solution to $A\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0} = (0,0)$.
Ex 5: $C(I) = \mathbf{R}^2$. $C(A)$ is the line of all multiples $c(1,2)$ in $\mathbf{R}^2$ because $(2,4) = 2(1,2)$. $C(B) = \mathbf{R}^2$; for $\boldsymbol{b} = (5,4)$, $\boldsymbol{x} = (0,1,1)$ and $\boldsymbol{x} = (2,0,1)$ both solve $B\boldsymbol{x} = \boldsymbol{b}$.

3.1 A: Put $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ into the columns of $A = \begin{bmatrix} \boldsymbol{b}_1 & \boldsymbol{b}_2 \end{bmatrix}$. Then $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \boldsymbol{b}_1$ and $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \boldsymbol{b}_2$. This construction works if and only if $\boldsymbol{b}_3$ is **not** a linear combination of $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$. If $\boldsymbol{b}_3 = c_1\boldsymbol{b}_1 + c_2\boldsymbol{b}_2$, then every subspace containing $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ must also contain $\boldsymbol{b}_3$, so no such matrix A exists.

3.1 B: Subspaces S and SS: For $\mathbf{V}_1$: $\mathbf{S}$ = combinations of $(1,1,0,0)$ and $(1,1,1,0)$; $\mathbf{SS}$ = multiples $(c,c,0,0)$. For $\mathbf{V}_2$: $\mathbf{S}$ = line through $(1,-1,1)$; $\mathbf{SS} = \mathbf{Z} = \{(0,0,0)\}$. For $\mathbf{V}_3$: $\mathbf{S}$ = diagonal matrices $\mathbf{D}$; $\mathbf{SS}$ = multiples cI. For $\mathbf{V}_4$: $\mathbf{S}$ = quadratic polynomials $a + bx + cx^2$; $\mathbf{SS}$ = linear polynomials $a + bx$.

Two descriptions of each V: $\mathbf{V}_1$: all combinations of $(1,1,0,0), (1,1,1,0), (1,1,1,1) \iff$ all solutions in $\mathbf{R}^4$ of $v_1 - v_2 = 0$. $\mathbf{V}_2$: all combinations of $(1,0,-1)$ and $(1,-1,1) \iff$ all solutions in $\mathbf{R}^3$ of $v_1 + 2v_2 + v_3 = 0$. $\mathbf{V}_3$: all combinations of $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \iff$ all $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfying $b - c = 0$. $\mathbf{V}_4$: all combinations of $1, x, x^2, x^3 \iff$ all solutions to $\mathrm{d}^4y/\mathrm{d}x^4 = 0$.

2.3 Problem Set 3.1 (Verbatim Questions and Solutions)

The first problems 1–8 are about vector spaces in general.

1. Suppose $(x_1, x_2) + (y_1, y_2)$ is defined to be $(x_1 + y_2, x_2 + y_1)$. With the usual multiplication $c\boldsymbol{x} = (cx_1, cx_2)$, which of the eight conditions are not satisfied?
2. Suppose the multiplication $c\boldsymbol{x}$ is defined to produce $(cx_1, 0)$ instead of (cx_1, cx_2). With the usual addition in $\mathbf{R}^2$, are the eight conditions satisfied?
3. **(a)** Which rules are broken if we keep only the positive numbers $x > 0$ in $\mathbf{R}^1$? Every c must be allowed. The half-line is not a subspace.
   (b) The positive numbers with $x + y$ and cx redefined to equal the usual xy and x^c *do* satisfy the eight rules. Test rule 7 when $c = 3, x = 2, y = 1$. (Then $x + y = 2$ and $cx = 8$.) Which number acts as the "zero vector"?
4. The matrix $A = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$ is a "vector" in the space $\mathbf{M}$ of all 2 by 2 matrices. Write down the zero vector in this space, the vector $\frac{1}{2}A$, and the vector $-A$. What matrices are in the smallest subspace containing A?
5. **(a)** Describe a subspace of $\mathbf{M}$ that contains $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ but not $B = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$.
   (b) If a subspace of $\mathbf{M}$ contains A and B, must it contain I?
   (c) Describe a subspace of $\mathbf{M}$ that contains no nonzero diagonal matrices.
6. The functions $f(x) = x^2$ and $g(x) = 5x$ are "vectors" in $\mathbf{F}$. This is the vector space of all real functions. (The functions are defined for $-\infty < x < \infty$.) The combination $3f(x) - 4g(x)$ is the function $h(x) =$ _____.
7. Which rule is broken if multiplying $f(x)$ by c gives the function $f(cx)$? Keep the usual addition $f(x) + g(x)$.
8. If the sum of the "vectors" $f(x)$ and $g(x)$ is defined to be the function $f(g(x))$, then the "zero vector" is $g(x) = x$. Keep the usual scalar multiplication $cf(x)$ and find two rules that are broken.

Questions 9–18 are about the "subspace requirements": $x + y$ and cx (and then all linear combinations $cx + dy$) stay in the subspace.

9. One requirement can be met while the other fails. Show this by finding
   (a) A set of vectors in $\mathbf{R}^2$ for which $\boldsymbol{x} + \boldsymbol{y}$ stays in the set but $\frac{1}{2}\boldsymbol{x}$ may be outside.
   (b) A set of vectors in $\mathbf{R}^2$ (other than two quarter-planes) for which every $c\boldsymbol{x}$ stays in the set but $\boldsymbol{x} + \boldsymbol{y}$ may be outside.
10. Which of the following subsets of $\mathbf{R}^3$ are actually subspaces?
    (a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$. **(b)** The plane of vectors with $b_1 = 1$.
    (c) The vectors with $b_1 b_2 b_3 = 0$. **(d)** All linear combinations of $\boldsymbol{v} = (1, 4, 0)$ and $\boldsymbol{w} = (2, 2, 2)$.
    (e) All vectors that satisfy $b_1 + b_2 + b_3 = 0$. **(f)** All vectors with $b_1 \le b_2 \le b_3$.
11. Describe the smallest subspace of the matrix space $\mathbf{M}$ that contains

$$\text{(a) } \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad \text{(b) } \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \qquad \text{(c) } \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

END-OF-PAGE SOLUTION Problem Set 3.1 (Problems 1–11)

1. Conditions **(1), (2), (3), (4)** fail: $(x_1 + y_2, x_2 + y_1) \neq (y_1 + x_2, y_2 + x_1)$; $\boldsymbol{x} + \boldsymbol{0} = (x_1, x_2)$ has no fixed zero vector since $(x_1 + 0_2, x_2 + 0_1) = (x_1, x_2)$ holds on the left but $\boldsymbol{0} + \boldsymbol{x} =$

$(x_2, x_1) \neq \boldsymbol{x}$.
2. Condition **(5)** fails: $1 \cdot (x_1, x_2) = (x_1, 0) \neq (x_1, x_2)$ when $x_2 \neq 0$.
3. (a) Scalar multiplication by $c \leq 0$ leaves the set, and rules **(3)** (no zero element 0) and **(4)** (no $-x$) fail. **(b)** Rule 7: $c \odot (x \oplus y) = (2 \cdot 1)^3 = 8$, and $(c \odot x) \oplus (c \odot y) = 2^3 \cdot 1^3 = 8$. The number **1** acts as the "zero vector" ($x \cdot 1 = x$).
4. Zero vector: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$; $\frac{1}{2}A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$; $-A = \begin{bmatrix} -2 & 2 \\ -2 & 2 \end{bmatrix}$. Smallest subspace: all multiples $cA = \begin{bmatrix} 2c & -2c \\ 2c & -2c \end{bmatrix}$.
5. (a) All matrices $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$. **(b) Yes**, $I = A - B$. **(c)** All matrices $\begin{bmatrix} 0 & b \\ c & 0 \end{bmatrix}$ with zero diagonal.
6. $\boldsymbol{h(x) = 3x^2 - 20x}$. **7.** Rule **(8)** fails: $(c_1 + c_2) \odot f = f((c_1 + c_2)x) \neq f(c_1 x) + f(c_2 x)$ for $f(x) = x^2$.
8. Rules **(1)** ($f(g(x)) \neq g(f(x))$), **(4)**, and **(7)** ($cf(g(x)) \neq (cf)(cg(x))$) fail.
9. (a) All integer vectors $(x_1, x_2) \in \mathbf{Z}^2$ (or first-quadrant vectors $x_1, x_2 \geq 0$). **(b)** The union of the x axis and y axis ($x_1 x_2 = 0$).
10. (a), (d), (e) are subspaces. (b) misses $(0, 0, 0)$; (c) is not closed under addition; (f) fails for $c = -1$.
11. (a) All $\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$. **(b)** All $\begin{bmatrix} c & c \\ 0 & 0 \end{bmatrix}$. **(c)** All diagonal matrices $\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$.

12. Let P be the plane in $\mathbf{R}^3$ with equation $x+y-2z=4$. The origin $(0,0,0)$ is not in P! Find two vectors in P and check that their sum is not in P.

13. Let $\mathbf{P}_0$ be the plane through $(0,0,0)$ parallel to the previous plane P. What is the equation for $\mathbf{P}_0$? Find two vectors in $\mathbf{P}_0$ and check that their sum is in $\mathbf{P}_0$.

14. The subspaces of $\mathbf{R}^3$ are planes, lines, $\mathbf{R}^3$ itself, or $\mathbf{Z}$ containing only $(0,0,0)$.
(a) Describe the three types of subspaces of $\mathbf{R}^2$.
(b) Describe all subspaces of $\mathbf{D}$, the space of 2 by 2 diagonal matrices.

15. **(a)** The intersection of two planes through $(0,0,0)$ is probably a _____ but it could be a _____. It can't be $\mathbf{Z}$!
(b) The intersection of a plane through $(0,0,0)$ with a line through $(0,0,0)$ is probably a _____ but it could be a _____.
(c) If $\mathbf{S}$ and $\mathbf{T}$ are subspaces of $\mathbf{R}^5$, prove that their intersection $\mathbf{S} \cap \mathbf{T}$ is a subspace of $\mathbf{R}^5$. Here $\mathbf{S} \cap \mathbf{T}$ consists of the vectors that lie in both subspaces. *Check the requirements on* $\boldsymbol{x}+\boldsymbol{y}$ *and* $c\boldsymbol{x}$.

16. Suppose $\mathbf{P}$ is a plane through $(0,0,0)$ and $\mathbf{L}$ is a line through $(0,0,0)$. The smallest vector space containing both $\mathbf{P}$ and $\mathbf{L}$ is either _____ or _____.

17. **(a)** Show that the set of *invertible* matrices in $\mathbf{M}$ is not a subspace.
(b) Show that the set of *singular* matrices in $\mathbf{M}$ is not a subspace.

18. True or false (check addition in each case by an example):
(a) The symmetric matrices in $\mathbf{M}$ (with $A^T = A$) form a subspace.
(b) The skew-symmetric matrices in $\mathbf{M}$ (with $A^T = -A$) form a subspace.
(c) The unsymmetric matrices in $\mathbf{M}$ (with $A^T \neq A$) form a subspace.

Questions 19–27 are about column spaces $C(A)$ and the equation $A\boldsymbol{x}=\boldsymbol{b}$.

19. Describe the column spaces (lines or planes) of these particular matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

20. For which right sides (find a condition on b_1, b_2, b_3) are these systems solvable?

$$\text{(a)} \begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \qquad \text{(b)} \begin{bmatrix} 1 & 4 \\ 2 & 9 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

END-OF-PAGE SOLUTION Problem Set 3.1 (Problems 12–20)

12. $(4,0,0)$ and $(0,4,0)$ are in P. Their sum $(4,4,0)$ gives $4+4-0=8 \neq 4$ (not in P).

13. Equation for $\mathbf{P}_0$: $x+y-2z=0$. Vectors $(1,1,1)$ and $(2,0,1)$ are in $\mathbf{P}_0$; their sum $(3,1,2)$ satisfies $3+1-4=0$.

14. **(a)** $\mathbf{R}^2$ itself, any line through $(0,0)$, or $\mathbf{Z} = \{(0,0)\}$. **(b)** $\mathbf{D}$ itself (dim 2), any line of multiples $c \begin{bmatrix} a_0 & 0 \\ 0 & d_0 \end{bmatrix}$ (dim 1), or the zero matrix $\mathbf{Z}$ (dim 0).

15. **(a) line**, **plane** (if the two planes coincide). **(b) point** $\mathbf{Z} = \{(0,0,0)\}$, **line** (if the line lies in the plane). **(c)** If $\boldsymbol{x},\boldsymbol{y} \in \mathbf{S} \cap \mathbf{T}$, then $\boldsymbol{x},\boldsymbol{y} \in \mathbf{S} \implies \boldsymbol{x}+\boldsymbol{y} \in \mathbf{S}, c\boldsymbol{x} \in \mathbf{S}$, and $\boldsymbol{x},\boldsymbol{y} \in \mathbf{T} \implies \boldsymbol{x}+\boldsymbol{y} \in \mathbf{T}, c\boldsymbol{x} \in \mathbf{T}$. Hence $\boldsymbol{x}+\boldsymbol{y}$ and $c\boldsymbol{x}$ lie in $\mathbf{S} \cap \mathbf{T}$.

16. Either $\mathbf{P}$ (if $\mathbf{L} \subset \mathbf{P}$) or $\mathbf{R}^3$ (if $\mathbf{L} \not\subset \mathbf{P}$).

17. (a) Zero matrix is not invertible (and $I + (-I) = 0$). **(b)** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ are singular, but their sum I is invertible.

18. (a) True: $(A+B)^T = A^T + B^T = A + B$. **(b) True:** $(A+B)^T = -(A+B)$. **(c) False:** $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is symmetric.

19. $C(A)$ is the x axis: $(x, 0, 0)$. $C(B)$ is the xy plane: $(x, y, 0)$. $C(C)$ is the line $(c, 2c, 0)$ in the xy plane.

20. (a) All columns are multiples of $(1, 2, -1)$, so $\mathbf{b_2} = \mathbf{2b_1}$ and $\mathbf{b_3} = -\mathbf{b_1}$. **(b)** Row 1 + Row 3 $= 0$, while Row 2 is independent, so the single condition is $\mathbf{b_1} + \mathbf{b_3} = \mathbf{0}$ (b_2 is free).

21. Adding row 1 of A to row 2 produces B. Adding column 1 to column 2 produces C. A combination of the columns of (B or C?) is also a combination of the columns of A. Which two matrices have the same column _____?

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

22. For which vectors (b_1, b_2, b_3) do these systems have a solution?

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

23. (Recommended) If we add an extra column $\boldsymbol{b}$ to a matrix A, then the column space gets larger unless _____. Give an example where the column space gets larger and an example where it doesn't. Why is $A\boldsymbol{x} = \boldsymbol{b}$ solvable exactly when the column space *doesn't* get larger—it is the same for A and $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$?

24. The columns of AB are combinations of the columns of A. This means: *The column space of AB is contained in (possibly equal to) the column space of A.* Give an example where the column spaces of A and AB are not equal.

25. Suppose $A\boldsymbol{x} = \boldsymbol{b}$ and $A\boldsymbol{y} = \boldsymbol{b}^*$ are both solvable. Then $A\boldsymbol{z} = \boldsymbol{b} + \boldsymbol{b}^*$ is solvable. What is $\boldsymbol{z}$? This translates into: *If $\boldsymbol{b}$ and $\boldsymbol{b}^*$ are in the column space $C(A)$, then $\boldsymbol{b} + \boldsymbol{b}^*$ is in $C(A)$.*

26. If A is any 5 by 5 invertible matrix, then its column space is _____. Why?

27. True or false (with a counterexample if false):

(a) The vectors $\boldsymbol{b}$ that are not in the column space $C(A)$ form a subspace.

(b) If $C(A)$ contains only the zero vector, then A is the zero matrix.

(c) The column space of $2A$ equals the column space of A.

(d) The column space of $A - I$ equals the column space of A (test this).

28. Construct a 3 by 3 matrix whose column space contains $(1,1,0)$ and $(1,0,1)$ but not $(1,1,1)$. Construct a 3 by 3 matrix whose column space is only a line.

29. If the 9 by 12 system $A\boldsymbol{x} = \boldsymbol{b}$ is solvable for *every* $\boldsymbol{b}$, then $C(A) =$ _____.

30. Challenge Problems: Suppose $\mathbf{S}$ and $\mathbf{T}$ are two subspaces of a vector space $\mathbf{V}$.

(a) Definition: The **sum** $\mathbf{S} + \mathbf{T}$ contains all sums $\boldsymbol{s} + \boldsymbol{t}$ of a vector $\boldsymbol{s}$ in $\mathbf{S}$ and a vector $\boldsymbol{t}$ in $\mathbf{T}$. Show that $\mathbf{S} + \mathbf{T}$ satisfies the requirements (addition and scalar multiplication) for a vector space.

(b) If $\mathbf{S}$ and $\mathbf{T}$ are lines in $\mathbf{R}^m$, what is the difference between $\mathbf{S} + \mathbf{T}$ and $\mathbf{S} \cup \mathbf{T}$? That union contains all vectors from $\mathbf{S}$ or $\mathbf{T}$ or both. Explain this statement: *The span of $\mathbf{S} \cup \mathbf{T}$ is $\mathbf{S} + \mathbf{T}$.*

31. If $\mathbf{S}$ is the column space of A and $\mathbf{T}$ is $C(B)$, then $\mathbf{S} + \mathbf{T}$ is the column space of what matrix M? The columns of A and B and M are all in $\mathbf{R}^m$. (I don't think $A + B$ is always a correct M.)

32. Show that the matrices A and $\begin{bmatrix} A & AB \end{bmatrix}$ (with extra columns) have the same column space. But find a square matrix with $C(A^2)$ smaller than $C(A)$. Important point: An n by n matrix has $C(A) = \mathbf{R}^n$ exactly when A is an _____ matrix.

END-OF-PAGE SOLUTION Problem Set 3.1 (Problems 21–32)

21. Columns of $\mathbf{C}$ are combinations of columns of A. A and C have the same column **space** (line through $(1,2)$).

22. Matrix 1: **all** $\boldsymbol{b} \in \mathbf{R}^3$. Matrix 2: $\mathbf{b_3} = \mathbf{0}$. Matrix 3: $\mathbf{b_2} = \mathbf{b_3}$.

23. Unless $\boldsymbol{b}$ **is already in** $C(A)$. Larger: $A = \begin{bmatrix}1\\0\end{bmatrix}, \boldsymbol{b} = \begin{bmatrix}0\\1\end{bmatrix}$. Same: $A = \begin{bmatrix}1\\0\end{bmatrix}, \boldsymbol{b} = \begin{bmatrix}2\\0\end{bmatrix}$. $A\boldsymbol{x} = \boldsymbol{b}$ means $\boldsymbol{b}$ is a combination of the columns of A, so $\boldsymbol{b}$ adds no new direction to $C(A)$.
24. Take $A = I$ and $B = 0$: $C(A) = \mathbf{R}^n$ while $C(AB) = \{\mathbf{0}\}$. **25.** $\boldsymbol{z} = \mathbf{x} + \mathbf{y}$ because $A(\boldsymbol{x} + \boldsymbol{y}) = A\boldsymbol{x} + A\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{b}^*$.
26. $\mathbf{R}^5$, because $A\boldsymbol{x} = \boldsymbol{b}$ is solved by $\boldsymbol{x} = A^{-1}\boldsymbol{b}$ for every $\boldsymbol{b} \in \mathbf{R}^5$.
27. (a) False ($\mathbf{0}$ is not in that set). **(b) True** (every column $A\boldsymbol{e}_j = \mathbf{0}$). **(c) True** ($2A\boldsymbol{x} = A(2\boldsymbol{x})$). **(d) False** (take $A = I$, $C(I - I) = \{\mathbf{0}\} \neq C(I)$).
28. Plane: $A = \begin{bmatrix}1&1&0\\1&0&0\\0&1&0\end{bmatrix}$. Line: $A = \begin{bmatrix}1&0&0\\1&0&0\\1&0&0\end{bmatrix}$. **29.** $C(A) = \mathbf{R}^9$.
30. (a) $(\boldsymbol{s}_1 + \boldsymbol{t}_1) + (\boldsymbol{s}_2 + \boldsymbol{t}_2) = (\boldsymbol{s}_1 + \boldsymbol{s}_2) + (\boldsymbol{t}_1 + \boldsymbol{t}_2) \in \mathbf{S} + \mathbf{T}$ and $c(\boldsymbol{s} + \boldsymbol{t}) = c\boldsymbol{s} + c\boldsymbol{t} \in \mathbf{S} + \mathbf{T}$. **(b)** $\mathbf{S} + \mathbf{T}$ is the entire plane spanned by the two lines, whereas $\mathbf{S} \cup \mathbf{T}$ is only the two lines themselves.
31. $M = \begin{bmatrix}A & B\end{bmatrix}$ (place the columns of A and B side by side).
32. Every column of AB is already a combination of the columns of A, so $C(\begin{bmatrix}A & AB\end{bmatrix}) = C(A)$. For $A = \begin{bmatrix}0&1\\0&0\end{bmatrix}$, $A^2 = \begin{bmatrix}0&0\\0&0\end{bmatrix}$, so $C(A^2) = \{\mathbf{0}\} \subsetneq C(A)$. An $n \times n$ matrix has $C(A) = \mathbf{R}^n$ exactly when A is an **invertible** matrix.

2.4 The Nullspace of A: Solving $Ax = 0$ (Section 3.2)

GOAL

Derive the subspace property of the nullspace $N(A)$, construct the special solutions of $A\boldsymbol{x} = \mathbf{0}$ from the echelon form U and reduced row echelon form R, and prove that $n > m$ guarantees nonzero solutions.

Step 1: Define the Nullspace $N(A)$ and prove it is a subspace of $\mathbf{R}^n$. For an $m \times n$ matrix A, the **nullspace** $N(A)$ consists of all solutions $\boldsymbol{x} \in \mathbf{R}^n$ to $A\boldsymbol{x} = \mathbf{0}$:

$$\boxed{N(A) = \{\boldsymbol{x} \in \mathbf{R}^n : A\boldsymbol{x} = \mathbf{0}\} \subseteq \mathbf{R}^n.} \tag{2.4}$$

If $\boldsymbol{x}, \boldsymbol{y} \in N(A)$, then $A(\boldsymbol{x}+\boldsymbol{y}) = A\boldsymbol{x} + A\boldsymbol{y} = \mathbf{0}+\mathbf{0} = \mathbf{0}$ and $A(c\boldsymbol{x}) = c(A\boldsymbol{x}) = \mathbf{0}$. Thus $N(A)$ is a subspace of $\mathbf{R}^n$.

Step 2: Forward Elimination to Echelon Form U and Reduced Form $R = \text{rref}(A)$. By **T4**, row operations do not change the solutions of $A\boldsymbol{x} = \mathbf{0}$, so $N(A) = N(U) = N(R)$:

1. **Echelon matrix U:** The first nonzero entry in each nonzero row is a **pivot**. Entries below every pivot are zero, forming a staircase pattern.
2. **Reduced row echelon form R:** Eliminate upward to produce zeros *above* every pivot, and divide each pivot row by its pivot so every pivot equals 1. The pivot columns of R contain an $r \times r$ identity matrix I.

Step 3: Separate Pivot Variables and Free Variables to Build Special Solutions. Suppose U and R have r pivots:

- The r columns with pivots correspond to **pivot variables**.
- The remaining $n - r$ columns without pivots correspond to **free variables**.

To find the $n - r$ **special solutions** $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_{n-r}$: set one free variable equal to 1, set all other free variables equal to 0, and solve $U\boldsymbol{x} = \mathbf{0}$ (or $R\boldsymbol{x} = \mathbf{0}$) for the r pivot variables. The complete solution to $A\boldsymbol{x} = \mathbf{0}$ is all linear combinations of the special solutions:

$$\boxed{\boldsymbol{x}_{\text{nullspace}} = c_1\boldsymbol{s}_1 + c_2\boldsymbol{s}_2 + \cdots + c_{n-r}\boldsymbol{s}_{n-r} = N\boldsymbol{c}.} \tag{2.5}$$

RESULT (Short Wide Matrices: More Unknowns than Equations, $n > m$)

An $m \times n$ matrix has at most m pivots ($r \leq m$). If $n > m$, there are at least $n - m \geq 1$ free variables. Therefore, **every system $A\boldsymbol{x} = \mathbf{0}$ with $n > m$ has nonzero solutions in $N(A)$.**

2.5 In-Text and Worked Examples (Section 3.2)

BOOK EXAMPLE 3.2 In-Text 1–4 (p. 132–137) Special Solutions and Reduced Form R

Ex 1: Find the nullspace and two special solutions for $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ $(x + 2y + 3z = 0)$.

Ex 2: Find the nullspace and special solution for $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$.

Ex 3: Describe the nullspaces of $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$, $B = \begin{bmatrix} A \\ 2A \end{bmatrix}$, and $C = \begin{bmatrix} A & 2A \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 3 & 8 & 6 & 16 \end{bmatrix}$. Find U, R, and the special solutions for C.

Ex 4: Find the nullspace, $R = \text{rref}(U)$, and special solution for $U = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 0 & 9 \end{bmatrix}$.

BOOK EXAMPLE 3.2 A Constructing R and A from Given Special Solutions

Create a 3 by 4 matrix whose special solutions to $A\boldsymbol{x} = \boldsymbol{0}$ are $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$:

$$\boldsymbol{s}_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \boldsymbol{s}_2 = \begin{bmatrix} -2 \\ 0 \\ -6 \\ 1 \end{bmatrix} \qquad \begin{array}{l} \text{pivot columns 1 and 3} \\ \text{free variables } x_2 \text{ and } x_4 \end{array}$$

You could create the matrix A in row reduced form R. Then describe *all* possible matrices A with the required nullspace $N(A) =$ all combinations of $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$.

BOOK EXAMPLE 3.2 B Special Solutions and Complete Solution to $Ax = 0$

Find the special solutions and describe the *complete solution* to $A\boldsymbol{x} = \boldsymbol{0}$ for

$$A_1 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix} \qquad A_3 = \begin{bmatrix} A_2 & A_2 \end{bmatrix}.$$

Which are the pivot columns? Which are the free variables? What is R in each case?

END-OF-PAGE SOLUTION Section 3.2 In-Text 1–4 & Worked Examples 3.2 A, 3.2 B

Ex 1: Column 1 has the pivot; y, z are free. Set $(y, z) = (1, 0) \implies \boldsymbol{s}_1 = (-2, 1, 0)$. Set $(y, z) = (0, 1) \implies \boldsymbol{s}_2 = (-3, 0, 1)$. $N(A)$ is the plane $c_1\boldsymbol{s}_1 + c_2\boldsymbol{s}_2$. **Ex 2:** $x_1 + 2x_2 = 0 \implies \boldsymbol{s} = (-2, 1)$, $N(A) = \{c(-2, 1)\}$. **Ex 3:** $N(A) = N(B) = \mathbf{Z} = \{(0, 0)\}$ (both columns are pivot columns). For C, $U = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 0 & 2 & 0 & 4 \end{bmatrix}$ and $R = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$. Free variables x_3, x_4 give $\boldsymbol{s}_1 = (-2, 0, 1, 0)$ and $\boldsymbol{s}_2 = (0, -2, 0, 1)$. **Ex 4:** $R = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; x_2 is free, giving $\boldsymbol{s} = (-5, 1, 0)$ and $N(U) = \{x_2(-5, 1, 0)\}$.

3.2 A: Pivots $= 1$ sit in columns 1 and 3, and row 3 is all zeros. Reverse the signs of the pivot entries in $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$ to fill free columns 2 and 4 of R:

$$R = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{satisfies} \quad R\boldsymbol{s}_1 = \boldsymbol{0} \text{ and } R\boldsymbol{s}_2 = \boldsymbol{0}.$$

All 3×4 matrices A with this nullspace have the form $A = ER$ where E is any invertible 3×3 matrix.

3.2 B: For A_1: No pivot columns; all 4 variables are free; $R_1 = A_1$. The 4 special solutions are the columns of I_4, and $N(A_1) = \mathbf{R}^4$. **For A_2:** Pivot column 1; free variable x_2; $R_2 = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$;

$\boldsymbol{s} = (-2, 1)$; complete solution $\boldsymbol{x} = c(-2, 1)$. **For A_3:** Pivot column 1; free variables x_2, x_3, x_4; $R_3 = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$; special solutions $\boldsymbol{s}_1 = (-2, 1, 0, 0)$, $\boldsymbol{s}_2 = (-1, 0, 1, 0)$, $\boldsymbol{s}_3 = (-2, 0, 0, 1)$.

2.6 Problem Set 3.2 (Verbatim Questions and Solutions)

Questions 1–4 and 5–8 are about the matrices in Problems 1 and 5.

1. Reduce these matrices to their ordinary echelon forms U:

$$\text{(a) } A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \qquad \text{(b) } B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{bmatrix}.$$

Which are the free variables and which are the pivot variables?

2. For the matrices in Problem 1, find a special solution for each free variable. (Set the free variable to 1. Set the other free variables to zero.)

3. By combining the special solutions in Problem 2, describe every solution to $A\boldsymbol{x} = \boldsymbol{0}$ and $B\boldsymbol{x} = \boldsymbol{0}$. The nullspace contains only $\boldsymbol{x} = \boldsymbol{0}$ when there are no _____.

4. By further row operations on each U in Problem 1, find the reduced echelon form R. *True or false:* The nullspace of R equals the nullspace of U.

5. By row operations reduce each matrix to its echelon form U. Write down a 2 by 2 lower triangular L such that $B = LU$.

$$\text{(a) } A = \begin{bmatrix} -1 & 3 & 5 \\ -2 & 6 & 10 \end{bmatrix} \qquad \text{(b) } B = \begin{bmatrix} -1 & 3 & 5 \\ -2 & 6 & 7 \end{bmatrix}.$$

6. For the same A and B, find the special solutions to $A\boldsymbol{x} = \boldsymbol{0}$ and $B\boldsymbol{x} = \boldsymbol{0}$. For an m by n matrix, the number of pivot variables plus the number of free variables is _____.

7. In Problem 5, describe the nullspaces of A and B in two ways. Give the equations for the plane or the line, and give all vectors $\boldsymbol{x}$ that satisfy those equations as combinations of the special solutions.

8. Reduce the echelon forms U in Problem 5 to R. For each R draw a box around the identity matrix that is in the pivot rows and pivot columns.

Questions 9–17 are about free variables and pivot variables.

9. True or false (with reason if true or example to show it is false):
(a) A square matrix has no free variables. **(b)** An invertible matrix has no free variables.
(c) An m by n matrix has no more than n pivot variables. **(d)** An m by n matrix has no more than m pivot variables.

10. Construct 3 by 3 matrices A to satisfy these requirements (if possible):
(a) A has no zero entries but $U = I$. **(b)** A has no zero entries but $R = I$.
(c) A has no zero entries but $R = U$. **(d)** $A = U = 2R$.

11. Put as many 1's as possible in a 4 by 7 echelon matrix U whose pivot columns are
(a) $2, 4, 5$ **(b)** $1, 3, 6, 7$ **(c)** 4 and 6.

12. Put as many 1's as possible in a 4 by 8 *reduced* echelon matrix R so that the free columns are
(a) $2, 4, 5, 6$ **(b)** $1, 3, 6, 7, 8$.

13. Suppose column 4 of a 3 by 5 matrix is all zero. Then x_4 is certainly a _____ variable. The special solution for this variable is the vector $\boldsymbol{x} =$ _____.

14. Suppose the first and last columns of a 3 by 5 matrix are the same (not zero). Then _____ is a free variable. Find the special solution for this variable.

15. Suppose an m by n matrix has r pivots. The number of special solutions is _____. The nullspace contains only $\boldsymbol{x} = \boldsymbol{0}$ when $r =$ _____. The column space is all of $\mathbf{R}^m$ when $r =$ _____.

16. The nullspace of a 5 by 5 matrix contains only $\boldsymbol{x} = \mathbf{0}$ when the matrix has _____ pivots. The column space is $\mathbf{R}^5$ when there are _____ pivots. Explain why.

17. The equation $x - 3y - z = 0$ determines a plane in $\mathbf{R}^3$. What is the matrix A in this equation? Which are the free variables? The special solutions are $(3, 1, 0)$ and _____.

END-OF-PAGE SOLUTION Problem Set 3.2 (Problems 1–17)

1. **(a)** $U_A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$: pivot variables x_1, x_3; free variables x_2, x_4, x_5. **(b)** $U_B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix}$: pivot variables x_1, x_2; free variable x_3.

2. **(a)** $\boldsymbol{s}_1 = (-2, 1, 0, 0, 0)$, $\boldsymbol{s}_2 = (0, 0, -2, 1, 0)$, $\boldsymbol{s}_3 = (0, 0, -3, 0, 1)$. **(b)** $\boldsymbol{s} = (1, -1, 1)$.

3. $A\boldsymbol{x} = \mathbf{0} : \boldsymbol{x} = x_2\boldsymbol{s}_1 + x_4\boldsymbol{s}_2 + x_5\boldsymbol{s}_3$; $B\boldsymbol{x} = \mathbf{0} : \boldsymbol{x} = x_3(1, -1, 1)$. Only $\boldsymbol{x} = \mathbf{0}$ when there are no **free variables**.

4. $R_A = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, $R_B = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. **True**: $N(R) = N(U)$.

5. $U_A = \begin{bmatrix} -1 & 3 & 5 \\ 0 & 0 & 0 \end{bmatrix}$, $U_B = \begin{bmatrix} -1 & 3 & 5 \\ 0 & 0 & -3 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$.

6. For A: $\boldsymbol{s}_1 = (3, 1, 0), \boldsymbol{s}_2 = (5, 0, 1)$. For B: $\boldsymbol{s} = (3, 1, 0)$. Pivot + free variables = **n**.

7. $N(A)$: plane $-x_1 + 3x_2 + 5x_3 = 0$, $\boldsymbol{x} = x_2(3, 1, 0) + x_3(5, 0, 1)$. $N(B)$: line $-x_1 + 3x_2 = 0, x_3 = 0$, $\boldsymbol{x} = x_2(3, 1, 0)$.

8. $R_A = \begin{bmatrix} \boxed{1} & -3 & -5 \\ 0 & 0 & 0 \end{bmatrix}$, $R_B = \begin{bmatrix} \boxed{1} & -3 & \boxed{0} \\ \boxed{0} & 0 & \boxed{1} \end{bmatrix}$.

9. **(a) False** (singular square matrix). **(b) True**. **(c) True**. **(d) True**.

10. **(a)** Impossible ($U = I \implies a_{21} = 0$? Wait: L can be non-zero below diagonal! E.g. $A = LI = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ still has zeros above diagonal; if $U = I$, $A = LU = L$ is lower triangular so $a_{12} = 0$, **impossible**). **(b)** Possible: $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ has $R = I$. **(c)** Impossible (U has zeros below row 1). **(d)** $A = U = 2I$ (or $A = U = R = 0$).

11. **(a)** Row 1 has 1s in cols 2–7 (6 ones), Row 2 in cols 4–7 (4 ones), Row 3 in cols 5–7 (3 ones), Row 4 all 0s (total 13 ones). **(b)** Cols 1–7 (7), 3–7 (5), 6–7 (2), 7 (1) = 15 ones. **(c)** Cols 4–7 (4), 6–7 (2) = 6 ones.

12. **(a)** Pivot cols 1, 3, 7, 8. Put 1s in pivots (4) plus free cols above/after pivots: col 2 has 1 in row 1 (1), cols 4,5,6 have 1s in rows 1,2 (6). Total = 11 ones. **(b)** Pivot cols 2, 4, 5 (max 3 pivots in 4 rows): pivots (3) + col 3 in row 1 (1) + cols 6,7,8 in rows 1,2,3 (9) = 13 ones.

13. **free**; $\boldsymbol{x} = (\mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{1}, \mathbf{0})$. **14.** $\mathbf{x_5}$ is free; $\boldsymbol{x} = (\mathbf{-1}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{1})$.

15. $\mathbf{n} - \mathbf{r}$; $r = \mathbf{n}$; $r = \mathbf{m}$. **16.** **5** pivots; **5** pivots ($R = I_5$).

17. $A = \begin{bmatrix} 1 & -3 & -1 \end{bmatrix}$; free variables y, z; second special solution is $(\mathbf{1}, \mathbf{0}, \mathbf{1})$.

18. (Recommended) The plane $x - 3y - z = 12$ is parallel to the plane $x - 3y - z = 0$ in Problem 17. One particular point on this plane is $(12, 0, 0)$. All points on the plane have the form (fill in the first components)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

19. Prove that U and $A = LU$ have the same nullspace when L is invertible:

$$\text{If } U\boldsymbol{x} = \boldsymbol{0} \text{ then } LU\boldsymbol{x} = \boldsymbol{0}. \quad \text{If } LU\boldsymbol{x} = \boldsymbol{0}, \text{ how do you know } U\boldsymbol{x} = \boldsymbol{0}?$$

20. Suppose column 1 + column 3 + column 5 = **0** in a 4 by 5 matrix with four pivots. Which column is sure to have no pivot (and which variable is free)? What is the special solution? What is the nullspace?

Questions 21–28 ask for matrices (if possible) with specific properties.

21. Construct a matrix whose nullspace consists of all combinations of $(2, 2, 1, 0)$ and $(3, 1, 0, 1)$.

22. Construct a matrix whose nullspace consists of all multiples of $(4, 3, 2, 1)$.

23. Construct a matrix whose column space contains $(1, 1, 5)$ and $(0, 3, 1)$ and whose nullspace contains $(1, 1, 2)$.

24. Construct a matrix whose column space contains $(1, 1, 0)$ and $(0, 1, 1)$ and whose nullspace contains $(1, 0, 1)$ and $(0, 0, 1)$.

25. Construct a matrix whose column space contains $(1, 1, 1)$ and whose nullspace is the line of multiples of $(1, 1, 1, 1)$.

26. Construct a 2 by 2 matrix whose nullspace equals its column space. This is possible.

27. Why does no 3 by 3 matrix have a nullspace that equals its column space?

28. If $AB = 0$ then the column space of B is contained in the _____ of A. Give an example of A and B.

29. The reduced form R of a 3 by 3 matrix with randomly chosen entries is almost sure to be _____. What R is virtually certain if the random A is 4 by 3?

30. Show by example that these three statements are generally *false*:
(a) A and A^T have the same nullspace. **(b)** A and A^T have the same free variables.
(c) If R is the reduced form rref(A) then R^T is rref(A^T).

31. If the nullspace of A consists of all multiples of $\boldsymbol{x} = (2, 1, 0, 1)$, how many pivots appear in U? What is R?

32. If the special solutions to $R\boldsymbol{x} = \boldsymbol{0}$ are in the columns of these N, go backward to find the nonzero rows of the reduced matrices R:

$$N = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} \\ \\ \end{bmatrix} \text{ (empty 3 by 1).}$$

33. **(a)** What are the five 2 by 2 reduced echelon matrices R whose entries are all 0's and 1's?
(b) What are the eight 1 by 3 matrices containing only 0's and 1's? Are all eight of them reduced echelon matrices R?

34. Explain why A and $-A$ always have the same reduced echelon form R.

35. **Challenge Problems:** If A is 4 by 4 and invertible, describe all vectors in the nullspace of the 4 by 8 matrix $B = \begin{bmatrix} A & A \end{bmatrix}$.

36. How is the nullspace $N(C)$ related to the spaces $N(A)$ and $N(B)$, if $C = \begin{bmatrix} A \\ B \end{bmatrix}$?

37. Kirchhoff's Law says that *current in* = *current out* at every node. This network (p. 143) has six currents $y_1, \ldots, y_6$. Find the four equations $A\boldsymbol{y} = \mathbf{0}$ for Kirchhoff's Law at the four nodes. Find three special solutions in the nullspace of A.

END-OF-PAGE SOLUTION Problem Set 3.2 (Problems 18–37)

18. First components are **12**, **3**, and **1**: $\boldsymbol{x} = (12,0,0) + y(3,1,0) + z(1,0,1)$.

19. Multiply $LU\boldsymbol{x} = \mathbf{0}$ by L^{-1} on the left: $L^{-1}LU\boldsymbol{x} = L^{-1}\mathbf{0} \implies U\boldsymbol{x} = \mathbf{0}$.

20. Column 5 has no pivot (x_5 is free). Special solution: $\boldsymbol{s} = (-1,0,-1,0,1)$. $N(A) = \{c(-1,0,-1,0,1)\}$.

21. $A = \begin{bmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & -2 & -1 \end{bmatrix}$. **22.** $A = \begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$. **23.** $A = \begin{bmatrix} 1 & 0 & -1/2 \\ 1 & 3 & -2 \\ 5 & 1 & -3 \end{bmatrix}$.

24. Impossible: $(1,0,1)$ and $(0,0,1)$ in $N(A)$ imply $(1,0,0) \in N(A)$, so col $1 = \mathbf{0}$ and col $3 = \mathbf{0}$, leaving only 1 nonzero column (cannot span 2 independent vectors $(1,1,0)$ and $(0,1,1)$).

25. $A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}$. **26.** $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has $C(A) = N(A) = \{c(1,0)\}$.

27. Because $\dim C(A) + \dim N(A) = r + (3-r) = 3$ is odd, we cannot have $r = 3 - r$.

28. Contained in the **nullspace** of A. Example: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$. **29.** $R = I_3$; for 4×3, $R = \begin{bmatrix} I_3 \\ 0 \end{bmatrix}$.

30. Take $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$: $N(A) = \{c(1,0)\}$ (x_1 free), $N(A^T) = \{c(0,1)\}$ (x_2 free), and $R^T = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \neq \text{rref}(A^T) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

31. $4 - 1 = \mathbf{3}$ pivots. $R = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. **32.** $R_1 = \begin{bmatrix} 1 & -2 & -3 \end{bmatrix}$; $R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; $R_3 = I_3$.

33. **(a)** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. **(b)** Only 5 of the 8 are RREF ($\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$ and four starting with 1: $\begin{bmatrix} 1 & a & b \end{bmatrix}$; $\begin{bmatrix} 0 & 2 \ldots \end{bmatrix}$ no, all 8 have first nonzero entry equal to 1, so **all 8** are in RREF!).

34. Multiplying $-A$ by -1 (an elementary row scaling) yields A, so $\text{rref}(-A) = \text{rref}(A)$.

35. $B \begin{bmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \end{bmatrix} = A(\boldsymbol{x}_1 + \boldsymbol{x}_2) = \mathbf{0} \iff \boldsymbol{x}_1 = -\boldsymbol{x}_2$, so all vectors in $N(B)$ have the form $(\boldsymbol{x}, -\boldsymbol{x})$ for $\boldsymbol{x} \in \mathbf{R}^4$.

36. $N(C) = N(A) \cap N(B)$.

37. Node 1: $-y_1 + y_3 - y_4 = 0$; Node 2: $y_1 - y_2 + y_5 = 0$; Node 3: $y_2 - y_3 - y_6 = 0$; Node 4: $y_4 - y_5 + y_6 = 0$. Three loop special solutions: $\boldsymbol{s}_1 = (1,0,1,0,-1,1)$, $\boldsymbol{s}_2 = (1,1,0,0,0,1)$, $\boldsymbol{s}_3 = (0,1,1,1,1,0)$ (or standard free-variable choices for y_3, y_5, y_6: $(-1,0,1,0,0,0)$ wait:

reduce $A = \begin{bmatrix} -1 & 0 & 1 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{bmatrix} \to R = \begin{bmatrix} 1 & 0 & -1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, free y_3, y_5, y_6 give $\boldsymbol{s}_1 = (1,1,1,0,0,0)$, $\boldsymbol{s}_2 = (1,0,0,1,1,0)$, $\boldsymbol{s}_3 = (-1,1,0,-1,0,1)$).

2.7 The Rank and the Row Reduced Form (Section 3.3)

GOAL

Define the rank r of a matrix, derive the Rank-One Factorization $A = \boldsymbol{u}\boldsymbol{v}^T$, and derive the block formula $N = \begin{bmatrix} -F \\ I \end{bmatrix}$ for the nullspace matrix from $R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$.

Step 1: Define the Rank r and Pivot Columns.

$$\boxed{\text{The } \textbf{rank } r \text{ of } A \text{ is the number of pivots in } U \text{ and } R = \text{rref}(A).} \tag{2.6}$$

- The r **pivot columns** of A are the columns in the list `pivcol` that end up with pivots in U and R. They are *not* combinations of earlier columns.
- The $n - r$ **free columns** of A *are* combinations of earlier pivot columns. The entries in the free columns of R display the exact coefficients of those combinations.

Step 2: Derive the Rank-One Factorization $A = \boldsymbol{u}\boldsymbol{v}^T$. If A has rank $r = 1$, every row is a multiple of the first nonzero row $\boldsymbol{v}^T$, and every column is a multiple of the pivot column $\boldsymbol{u}$. Therefore:

$$\boxed{\text{Every rank-one matrix has the form } A = \boldsymbol{u}\boldsymbol{v}^T = (\text{column}) \text{ times } (\text{row}).} \tag{2.7}$$

More generally, a matrix of rank r is the sum of r rank-one matrices: $A = \boldsymbol{u}_1\boldsymbol{v}_1^T + \cdots + \boldsymbol{u}_r\boldsymbol{v}_r^T = (\text{pivot columns of } A)(\text{first } r \text{ rows of } R)$.

Step 3: Derive the Nullspace Matrix N from Block Reduced Form R. Suppose the r pivot columns come first in R. Then R and the nullspace matrix N have the block forms:

$$R = \begin{bmatrix} I_{r\times r} & F_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix} \Longrightarrow \boxed{N = \begin{bmatrix} -F \\ I_{n-r} \end{bmatrix} \quad \text{satisfies} \quad RN = \begin{bmatrix} -F+F \\ 0 \end{bmatrix} = 0.} \tag{2.8}$$

2.8 In-Text and Worked Examples (Section 3.3)

BOOK EXAMPLE 3.3 In-Text 1–2 (p. 144–148) Rank, Pivot Columns, and Nullspace Matrix N

In-Text (1): Find U, R, rank r, and special solutions for $A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix}$.

In-Text (2): Factor the rank-one matrix $A = \begin{bmatrix} 1 & 3 & 10 \\ 2 & 6 & 20 \\ 3 & 9 & 30 \end{bmatrix} = \boldsymbol{u}\boldsymbol{v}^T$ and find its special solutions.

In-Text (3): For $A = \begin{bmatrix} 1 & 3 & 0 & 2 & -1 \\ 0 & 0 & 1 & 4 & -3 \\ 1 & 3 & 1 & 6 & -4 \end{bmatrix}$, find R, the pivot columns, and the 5×3 nullspace matrix N.

BOOK EXAMPLE 3.3 A Second Difference Matrix without Boundary Conditions

Find the reduced echelon form of A. What is the rank? What is the special solution to $A\boldsymbol{x} = \mathbf{0}$?

$$\textbf{Second differences } -1, 2, -1 \qquad A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

BOOK EXAMPLE 3.3 B Rank-One and Rank-Two Matrix Factorizations

Factor these rank one matrices into $A = \boldsymbol{u}\boldsymbol{v}^T$ = column times row:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \qquad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ (find } d \text{ from } a, b, c \text{ if } a \neq 0).$$

Split this rank two matrix into $\boldsymbol{u}_1\boldsymbol{v}_1^T + \boldsymbol{u}_2\boldsymbol{v}_2^T$ = (3 by 2) times (2 by 4) using R:

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = E^{-1}R.$$

BOOK EXAMPLE 3.3 C Rank and Special Solutions Depending on a Parameter c

Find the row reduced form R and the rank r of A and B (those depend on c). Which are the pivot columns of A? What are the special solutions and the matrix N?

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & c \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} c & c \\ c & c \end{bmatrix}.$$

END-OF-PAGE SOLUTION Section 3.3 In-Text & Worked Examples 3.3 A, 3.3 B, 3.3 C

In-Text (1): $U = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $r = 2$. $\boldsymbol{s}_1 = (-2, 0, 1, 0), \boldsymbol{s}_2 = (-3, -1, 0, 1)$.

In-Text (2): $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 10 \end{bmatrix}$; $\boldsymbol{s}_1 = (-3, 1, 0), \boldsymbol{s}_2 = (-10, 0, 1)$. **In-Text (3):** $R = \begin{bmatrix} 1 & 3 & 0 & 2 & -1 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$; pivot cols 1, 3; $N = \begin{bmatrix} -3 & -2 & 1 \\ 1 & 0 & 0 \\ 0 & -4 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

3.3 A: Add row 1 to row 2, row 2 to row 3, row 3 to row 4 to get $U = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Eliminate upward to get $R = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The rank is $r = 3$, and the special solution is $\boldsymbol{s} = (1, 1, 1, 1)$.

3.3 B: $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$; $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 \\ c/a \end{bmatrix} \begin{bmatrix} a & b \end{bmatrix}$ requires $d = \frac{bc}{a}$ ($ad - bc = 0$). For the 3×4 matrix, $A = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$.

3.3 C: For A: if $c \neq 4$, $r = 2$ (pivot cols 1, 3), $R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $N = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$. If $c = 4$, $r = 1$ (pivot col 1), $R = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $N = \begin{bmatrix} -2 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. For B: if $c \neq 0$, $r = 1$, $R = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $N = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$; if $c = 0$, $r = 0$, $R = 0$, $N = I_2$.

2.9 Problem Set 3.3 (Verbatim Questions and Solutions)

1. Which of these rules gives a correct definition of the *rank* of A?
(a) The number of nonzero rows in R. **(b)** The number of columns minus the total number of rows.
(c) The number of columns minus the number of free columns. **(d)** The number of 1's in the matrix R.

2. Find the reduced row echelon forms R and the rank of these matrices:
(a) The 3 by 4 matrix with all entries equal to 4.
(b) The 3 by 4 matrix with $a_{ij} = i + j - 1$.
(c) The 3 by 4 matrix with $a_{ij} = (-1)^j$.

3. Find the reduced R for each of these (block) matrices:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 4 & 6 \end{bmatrix} \qquad B = \begin{bmatrix} A & A \end{bmatrix} \qquad C = \begin{bmatrix} A & A \\ A & 0 \end{bmatrix}.$$

4. Suppose all the pivot variables come *last* instead of first. Describe all four blocks in the reduced echelon form (the block B should be r by r):

$$R = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

What is the nullspace matrix N containing the special solutions?

5. (Silly problem) Describe all 2 by 3 matrices A_1 and A_2, with row echelon forms R_1 and R_2, such that $R_1 + R_2$ is the row echelon form of $A_1 + A_2$. Is it true that $R_1 = A_1$ and $R_2 = A_2$ in this case? Does $R_1 - R_2$ equal $\text{rref}(A_1 - A_2)$?

6. If A has r pivot columns, how do you know that A^T has r pivot columns? Give a 3 by 3 example with different column numbers in *pivcol* for A and A^T.

7. What are the special solutions to $R\boldsymbol{x} = \boldsymbol{0}$ and $\boldsymbol{y}^T R = \boldsymbol{0}$ for these R?

$$R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Problems 8–11 are about matrices of rank $r = 1$.

8. Fill out these matrices so that they have rank 1:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & & \\ 4 & & \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} & 9 & \\ 1 & & \\ 2 & 6 & -3 \end{bmatrix} \quad \text{and} \quad M = \begin{bmatrix} a & b \\ c & \end{bmatrix}.$$

9. If A is an m by n matrix with $r = 1$, its columns are multiples of one column and its rows are multiples of one row. The column space is a _____ in $\mathbf{R}^m$. The nullspace is a _____ in $\mathbf{R}^n$. The nullspace matrix N has shape _____.

10. Choose vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ so that $A = \boldsymbol{u}\boldsymbol{v}^T =$ column times row:

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ 4 & 8 & 8 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 2 & 6 & 4 \\ -1 & -1 & -3 & -2 \end{bmatrix}.$$

11. If A is a rank one matrix, the second row of U is _____. Do an example.

END-OF-PAGE SOLUTION Problem Set 3.3 (Problems 1–11)

1. **(a)** and **(c)** are correct (r = nonzero rows in $R = n - (n - r)$).

2. **(a)** $R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $r = 1$. **(b)** $R = \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $r = 2$. **(c)** $R = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $r = 1$.

3. $R_A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$; $R_B = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$; $R_C = \begin{bmatrix} R_A & 0 \\ 0 & R_A \end{bmatrix}$ with the two zero rows moved to the bottom (rows 5 and 6).

4. $R = \begin{bmatrix} F & I_r \\ 0 & 0 \end{bmatrix}$ (so $A = F, B = I_r, C = 0, D = 0$). Nullspace matrix: $N = \begin{bmatrix} I_{n-r} \\ -F \end{bmatrix}$.

5. For $R_1 + R_2$ to have pivots equal to 1, R_1 and R_2 cannot share any pivot column (otherwise $1 + 1 = 2$). $R_1 = A_1$ is not required; $R_1 - R_2 \neq \text{rref}(A_1 - A_2)$ because $-R_2$ has -1 in its pivots.

6. By the rank factorization $A =$ (pivot cols P)(nonzero rows of R), A^T has rank r as well. Example: $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ has $\text{pivcol}(A) = (2, 3)$, while A^T has $\text{pivcol}(A^T) = (1, 2)$.

7. Matrix 1: $R\boldsymbol{x} = \mathbf{0} \implies \boldsymbol{s}_1 = (-2, -4, 1, 0), \boldsymbol{s}_2 = (-3, -5, 0, 1)$; $\boldsymbol{y}^T R = \mathbf{0} \implies \boldsymbol{y} = (0, 0, 1)$. Matrix 2: $\boldsymbol{s}_1 = (1, 0, 0), \boldsymbol{s}_2 = (0, -2, 1)$; $\boldsymbol{y}_1 = (0, 1, 0), \boldsymbol{y}_2 = (0, 0, 1)$.

8. $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 4 & 8 & 16 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 9 & -9/2 \\ 1 & 3 & -3/2 \\ 2 & 6 & -3 \end{bmatrix}$, $M = \begin{bmatrix} a & b \\ c & bc/a \end{bmatrix}$.

9. **line** in $\mathbf{R}^m$; **hyperplane** (of dimension $n - 1$) in $\mathbf{R}^n$; shape $\mathbf{n}$ by $(\mathbf{n} - \mathbf{1})$.

10. First: $\boldsymbol{u} = (3, 1, 4), \boldsymbol{v} = (1, 2, 2)$. Second: $\boldsymbol{u} = (2, -1), \boldsymbol{v} = (1, 1, 3, 2)$. **11. all zeros** $\begin{bmatrix} 0 & \dots & 0 \end{bmatrix}$.

Problems 12–14 are about r by r invertible matrices inside A.

12. *If A has rank r, then it has an r by r submatrix S that is invertible.* Remove $m-r$ rows and $n-r$ columns to find an invertible submatrix S inside A, B, and C. You could keep the pivot rows and pivot columns:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \qquad C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

13. Suppose P contains only the r pivot columns of an m by n matrix. Explain why this m by r submatrix P has rank r.

14. Transpose P in Problem 13. Then find the r pivot columns of P^T. Transposing back, **this produces an r by r invertible submatrix S inside P and A**:

$$\text{For } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 4 & 7 \end{bmatrix} \text{ find } P \text{ (3 by 2) and then the invertible } S \text{ (2 by 2)}.$$

Problems 15–20 show that $\text{rank}(AB)$ is not greater than $\text{rank}(A)$ or $\text{rank}(B)$.

15. Find the ranks of AB and AC (rank one matrix times rank one matrix):

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1.5 & 6 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & b \\ c & bc \end{bmatrix}.$$

16. The rank one matrix $\boldsymbol{uv}^T$ times the rank one matrix $\boldsymbol{wz}^T$ is $\boldsymbol{uz}^T$ times the number _____. This product $\boldsymbol{uv}^T\boldsymbol{wz}^T$ also has rank one unless _____ $= 0$.

17. **(a)** Suppose column j of B is a combination of previous columns of B. Show that column j of AB is the same combination of previous columns of AB. Then AB cannot have new pivot columns, so $\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{B})$.

(b) Find A_1 and A_2 so that $\text{rank}(A_1B) = 1$ and $\text{rank}(A_2B) = 0$ for $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

18. Problem 17 proved that $\text{rank}(AB) \leq \text{rank}(B)$. Then the same reasoning gives $\text{rank}(\mathbf{B^T A^T}) \leq \text{rank}(\mathbf{A^T})$. How do you deduce that $\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{A})$?

19. (*Important*) Suppose A and B are n by n matrices, and $AB = I$. Prove from $\text{rank}(AB) \leq \text{rank}(A)$ that the rank of A is n. *So A is invertible and B must be its two-sided inverse (Section 2.5). Therefore $BA = I$ (which is not so obvious!).*

20. If A is 2 by 3 and B is 3 by 2 and $AB = I$, show from its rank that $BA \neq I$. Give an example of A and B with $AB = I$. For $m < n$, a right inverse is not a left inverse.

21. Suppose A and B have the *same* reduced row echelon form R.
(a) Show that A and B have the same nullspace and the same row space.
(b) We know $E_1A = R$ and $E_2B = R$. So A equals an _____ matrix times B.

22. Express A and then B as the sum of two rank one matrices:

$$\textbf{rank = 2} \qquad A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 8 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}.$$

23. Answer the same questions as in Worked Example **3.3 C** for

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 1 & c & 2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1-c & 2 \\ 0 & 2-c \end{bmatrix}.$$

24. What is the nullspace matrix N (containing the special solutions) for A, B, C?

$$A = \begin{bmatrix} I & I \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} I & I \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} I & I & I \end{bmatrix}.$$

25. *Neat fact* **Every m by n matrix of rank r reduces to (m by r) times (r by n):**

$$A = (\text{pivot columns of } A)\,(\text{first } r \text{ rows of } R) = \textbf{(COL)(ROW)}.$$

Write the 3 by 4 matrix A in equation (1) at the start of this section as the product of the 3 by 2 matrix from the pivot columns and the 2 by 4 matrix from R.

26. Challenge Problems: Suppose A is an m by n matrix of rank r. Its reduced echelon form is R. Describe *exactly* the matrix Z (its shape and all its entries) that comes from *transposing the reduced row echelon form of* R' (prime means transpose): $R = \text{rref}(A)$ and $Z = (\text{rref}(R'))'$.

27. Suppose R is m by n of rank r, with pivot columns first: $R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$.

(a) What are the shapes of those four blocks? **(b)** Find a *right-inverse* B with $RB = I$ if $r = m$.

(c) Find a *left-inverse* C with $CR = I$ if $r = n$. **(d)** What is the reduced row echelon form of R^T (with shapes)?

(e) What is the reduced row echelon form of $R^T R$ (with shapes)? Prove that $R^T R$ has the same nullspace as R.

28. Suppose you allow elementary *column* operations on A as well as elementary row operations (which get to R). What is the "row-and-column reduced form" for an m by n matrix of rank r?

END-OF-PAGE SOLUTION Problem Set 3.3 (Problems 12–28)

12. $S_A = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix}$ (rows 1,2; cols 1,3); $S_B = \begin{bmatrix} 1 \end{bmatrix}$ (row 1, col 1); $S_C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (rows 1,3; cols 2,3).

13. P keeps all r pivots during row reduction, so P has r pivots and rank r. **14.** $P = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 2 & 7 \end{bmatrix}$ (cols 1,3); $S = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ (rows 1,3).

15. $\text{rank}(AB) = 1$; $\text{rank}(AC) = 1$ unless $1 + 2c = 0$ ($c = -1/2$), where $AC = 0$ and $\text{rank}(AC) = 0$.

16. Number $\mathbf{v^T w}$; unless $\mathbf{v^T w} = 0$. **17. (a)** If $\boldsymbol{b}_j = \sum_{k<j} c_k \boldsymbol{b}_k$, then $A\boldsymbol{b}_j = \sum_{k<j} c_k (A\boldsymbol{b}_k)$. **(b)** $A_1 = I_2$, $A_2 = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$.

18. $\text{rank}(AB) = \text{rank}((AB)^T) = \text{rank}(B^T A^T) \le \text{rank}(A^T) = \text{rank}(A)$.

19. $n = \text{rank}(I) = \text{rank}(AB) \le \text{rank}(A) \le n \implies \text{rank}(A) = n \implies A^{-1}$ exists $\implies B = A^{-1} \implies BA = I$.

20. BA is 3×3 with $\text{rank}(BA) \le \text{rank}(A) \le 2 < 3 = \text{rank}(I_3)$, so $BA \neq I_3$. Example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$.

21. (a) $N(A) = N(R) = N(B)$ and row space $= C(R^T)$ for both. **(b)** $A = (E_1^{-1} E_2)B$, an

invertible matrix times B.

22. $A = \begin{bmatrix}1\\1\\1\end{bmatrix}\begin{bmatrix}1 & 1 & 0\end{bmatrix} + \begin{bmatrix}0\\4\\8\end{bmatrix}\begin{bmatrix}0 & 0 & 1\end{bmatrix}$; $B = \begin{bmatrix}2\\2\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}$.

23. For A: if $c \neq 1$, $r = 2$ (cols 1,2), $R = \begin{bmatrix}1 & 0 & 2 & 2\\0 & 1 & 0 & 0\\0 & 0 & 0 & 0\end{bmatrix}$; if $c = 1$, $r = 1$ (col 1), $R = \begin{bmatrix}1 & 1 & 2 & 2\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{bmatrix}$.

For B: if $c \neq 1, 2$, $r = 2$, $R = I_2$, $N =$ empty; if $c = 1$, $r = 1$ (col 2), $R = \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}$, $N = \begin{bmatrix}1\\0\end{bmatrix}$; if $c = 2$, $r = 1$ (col 1), $R = \begin{bmatrix}1 & -2\\0 & 0\end{bmatrix}$, $N = \begin{bmatrix}2\\1\end{bmatrix}$.

24. $N_A = N_B = \begin{bmatrix}-I\\I\end{bmatrix}$; $N_C = \begin{bmatrix}-I & -I\\I & 0\\0 & I\end{bmatrix}$. **25.** $A = \begin{bmatrix}1 & 1\\1 & 2\\1 & 3\end{bmatrix}\begin{bmatrix}1 & 0 & 2 & 3\\0 & 1 & 0 & 1\end{bmatrix}$.

26. Z is $m \times n$ with $\begin{bmatrix}I_r & 0\\0 & 0\end{bmatrix}$: the r pivot columns of R are reduced to I_r in the first r columns and all $n - r$ free columns become zero!

27. (a) I is $r \times r$, F is $r \times (n-r)$, bottom blocks are $(m-r) \times r$ and $(m-r) \times (n-r)$. **(b)** $B = \begin{bmatrix}I_m\\0\end{bmatrix}$. **(c)** $C = \begin{bmatrix}I_n & 0\end{bmatrix}$. **(d)** $\text{rref}(R^T) = \begin{bmatrix}I_r & 0\\0 & 0\end{bmatrix}$ $(n \times m)$. **(e)** $\text{rref}(R^T R) = \begin{bmatrix}I_r & F\\0 & 0\end{bmatrix}$ $(n \times n)$. Since $R^T R\boldsymbol{x} = \boldsymbol{0} \implies \boldsymbol{x}^T R^T R\boldsymbol{x} = \|R\boldsymbol{x}\|^2 = 0 \implies R\boldsymbol{x} = \boldsymbol{0}$, their nullspaces are identical.

28. $\begin{bmatrix}I_r & 0\\0 & 0\end{bmatrix}$ $(m \times n)$.

Module 2: Unique Solved Problem Archetypes & Formula Sheet

1. Subspace Verification and Column Space Solvability

- **Subspace Test (Ex 3.1 1–3, PS 3.1 #10):** Verify (0) $\mathbf{0} \in \mathbf{S}$ and (1) $c\boldsymbol{x} + d\boldsymbol{y} \in \mathbf{S}$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbf{S}$ and $c, d \in \mathbf{R}$. Any constant offset ($b_1 = 1$), product ($b_1 b_2 b_3 = 0$), or inequality ($b_1 \geq 0$) breaks subspace closure.
- **Column Space Condition:** $\boldsymbol{b} \in C(A) \iff A\boldsymbol{x} = \boldsymbol{b}$ has at least one solution $\boldsymbol{x} \iff \text{rank}(\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}) = \text{rank}(A)$.

2. Nullspace Matrix N, Rank r, and (COL)(ROW) Factorization

- **Special Solutions from R (Ex 3.2 A, 3.3 C):** If $R = \begin{bmatrix} I_r & F \\ 0 & 0 \end{bmatrix}$, then $N = \begin{bmatrix} -F \\ I_{n-r} \end{bmatrix}$ contains the $n - r$ special solutions in its columns, and $RN = 0$, $AN = 0$.
- **Rank Factorization (Ex 3.3 B, PS 3.3 #25):** Every $m \times n$ matrix of rank r factors as $A = PR_{1:r} = \sum_{k=1}^{r} \boldsymbol{u}_k \boldsymbol{v}_k^T$, where P ($m \times r$) holds the r pivot columns of A and $R_{1:r}$ ($r \times n$) holds the r nonzero rows of R.

MODULE 3

Complete Solution, Basis, Dimension, and the Four Subspaces

CHAPTER 3

Complete Solution $Ax = b$, Basis, and Four Subspaces (Sections 3.4–3.6)

Syllabus Coverage: Complete solution $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$ via augmented matrix $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix} \to \begin{bmatrix} R & \boldsymbol{d} \end{bmatrix}$; full column rank ($r = n$) and full row rank ($r = m$); linear independence, spanning sets, basis, and dimension of a vector space; bases for matrix and function spaces; dimensions and bases of the Four Fundamental Subspaces ($C(A^T), N(A), C(A), N(A^T)$) and the Fundamental Theorem of Linear Algebra, Part 1.

3.1 The Complete Solution to $Ax = b$ (Section 3.4)

GOAL

Derive the solvability conditions on $\boldsymbol{b}$ and construct the complete solution $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$ by reducing the augmented matrix $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ to $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix}$.

Step 1: Reduce the Augmented Matrix $\begin{bmatrix} A & b \end{bmatrix} \to \begin{bmatrix} U & c \end{bmatrix} \to \begin{bmatrix} R & d \end{bmatrix}$. Apply the exact same row operations to $\boldsymbol{b}$ that reduce A to R. The last $m - r$ rows of R are all zeros. For $R\boldsymbol{x} = \boldsymbol{d}$ to be consistent, the last $m - r$ components of $\boldsymbol{d}$ must equal zero ($0 = 0$).

Step 2: Construct One Particular Solution x_p. Set all $n - r$ *free variables* equal to 0. Because the r pivot columns of R contain the identity matrix I_r, the r *pivot variables* in $\boldsymbol{x}_p$ equal the first r entries of $\boldsymbol{d}$:

$$A\boldsymbol{x}_p = \boldsymbol{b} \quad \text{with} \quad (\boldsymbol{x}_p)_{\text{free}} = 0, \quad (\boldsymbol{x}_p)_{\text{pivot}} = \boldsymbol{d}_{1:r}. \tag{3.1}$$

Step 3: Add the Nullspace Solution x_n to Obtain the Complete Solution. If $A\boldsymbol{x}_p = \boldsymbol{b}$ and $A\boldsymbol{x}_n = \boldsymbol{0}$, then by linearity $A(\boldsymbol{x}_p + \boldsymbol{x}_n) = A\boldsymbol{x}_p + A\boldsymbol{x}_n = \boldsymbol{b} + \boldsymbol{0} = \boldsymbol{b}$. Conversely, if $A\boldsymbol{x} = \boldsymbol{b}$, then $A(\boldsymbol{x} - \boldsymbol{x}_p) = \boldsymbol{b} - \boldsymbol{b} = \boldsymbol{0}$, so $\boldsymbol{x} - \boldsymbol{x}_p = \boldsymbol{x}_n \in N(A)$.

$$\boxed{\boldsymbol{x}_{\text{complete}} = \boldsymbol{x}_p + \boldsymbol{x}_n = \boldsymbol{x}_p + c_1\boldsymbol{s}_1 + c_2\boldsymbol{s}_2 + \cdots + c_{n-r}\boldsymbol{s}_{n-r}.} \tag{3.2}$$

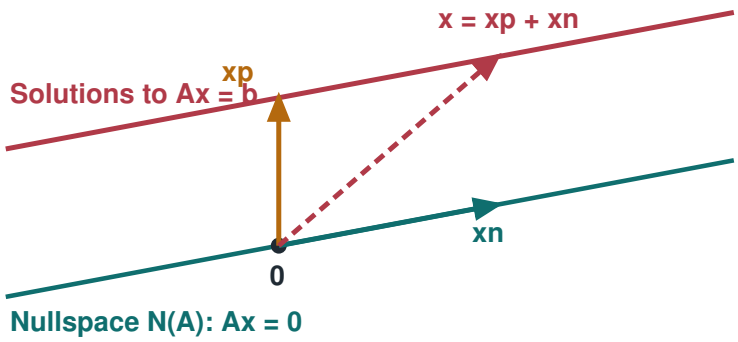


Fig. 3.1 Geometric meaning of $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$: the solutions to $A\boldsymbol{x} = \boldsymbol{b}$ form a shifted flat (line or plane) parallel to the nullspace $N(A)$, translated away from $\boldsymbol{0}$ by one particular solution $\boldsymbol{x}_p$.

RESULT (The Four Possibilities for Linear Equations Depend on the Rank r)

Case	Rank Condition	Shape / Type	Form of R	Number of Solutions to $A\boldsymbol{x} = \boldsymbol{b}$
1. Full rank	$r = m = n$	Square invertible	$R = \begin{bmatrix} I \end{bmatrix}$	**1** solution ($\boldsymbol{x} = A^{-1}\boldsymbol{b}$) for every $\boldsymbol{b}$
2. Full row rank	$r = m < n$	Short and wide	$R = \begin{bmatrix} I & F \end{bmatrix}$	**∞** solutions for every $\boldsymbol{b}$
3. Full column rank	$r = n < m$	Tall and thin	$R = \begin{bmatrix} I \\ 0 \end{bmatrix}$	**0 or 1** solution (unique if solvable)
4. Not full rank	$r < m$ and $r < n$	General rank-deficient	$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$	**0 or ∞** solutions

3.2 In-Text and Worked Examples (Section 3.4)

BOOK EXAMPLE 3.4 In-Text 0–2 (p. 155–158) Particular + Nullspace Solutions

In-Text 0 (p. 155): Solve $\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 1 & 3 & 1 & 6 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix}$ and find the solvability condition on (b_1, b_2, b_3).

Example 1 (p. 156): Find the condition on (b_1, b_2, b_3) for $A\boldsymbol{x} = \boldsymbol{b}$ to be solvable if $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ -2 & -3 \end{bmatrix}$. Find $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$.

Example 2 (p. 158): Find the complete solution to $x + y + z = 3$, $x + 2y - z = 4$ ($r = m = 2$).

BOOK EXAMPLE 3.4 A Connecting Elimination to Column Space, Nullspace, Rank, and Solvability

This question connects **elimination (pivot columns and back substitution)** to **column space-nullspace-rank-solvability** (the full picture). A has rank 2:

$$A\boldsymbol{x} = \boldsymbol{b} \quad \text{is} \quad \begin{aligned} x_1 + 2x_2 + 3x_3 + 5x_4 &= b_1 \\ 2x_1 + 4x_2 + 8x_3 + 12x_4 &= b_2 \\ 3x_1 + 6x_2 + 7x_3 + 13x_4 &= b_3 \end{aligned}$$

1. Reduce $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ to $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix}$, so that $A\boldsymbol{x} = \boldsymbol{b}$ becomes a triangular system $U\boldsymbol{x} = \boldsymbol{c}$.
2. Find the condition on b_1, b_2, b_3 for $A\boldsymbol{x} = \boldsymbol{b}$ to have a solution.
3. Describe the column space of A. *Which plane in* $\mathbf{R}^3$*?*
4. Describe the nullspace of A. *Which special solutions in* $\mathbf{R}^4$*?*
5. Find a particular solution to $A\boldsymbol{x} = (0, 6, -6)$ and then the complete solution.
6. Reduce $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix}$ to $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix}$: *Special solutions from* R*, particular solution from* $\boldsymbol{d}$.

BOOK EXAMPLE 3.4 B Deducing Matrix Shape and Rank from Solution Structure

If you have this information about the solutions to $A\boldsymbol{x} = \boldsymbol{b}$ **for a specific** $\boldsymbol{b}$, what does that tell you about the shape of A (and A itself)? And possibly about $\boldsymbol{b}$.

1. There is exactly one solution.
2. There are no solutions.
3. There are infinitely many solutions.
4. All solutions to $A\boldsymbol{x} = \boldsymbol{b}$ have the form $\boldsymbol{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.
2. All solutions to $A\boldsymbol{x} = \boldsymbol{b}$ have the form $\boldsymbol{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

BOOK EXAMPLE 3.4 C Complete Solution and Left-Nullspace Solvability Condition

Find the complete solution $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$ by forward elimination on $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$:

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 4 & 8 \\ 4 & 8 & 6 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 10 \end{bmatrix}.$$

Find numbers y_1, y_2, y_3 so that $y_1(\text{row } 1) + y_2(\text{row } 2) + y_3(\text{row } 3) = \text{zero row}$. Check that $\boldsymbol{b} = (4, 2, 10)$ satisfies the condition $y_1 b_1 + y_2 b_2 + y_3 b_3 = 0$.

END-OF-PAGE SOLUTION Section 3.4 In-Text & Worked Examples 3.4 A, 3.4 B, 3.4 C

In-Text 0: $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix} = \begin{bmatrix} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$; condition $b_3 - b_1 - b_2 = 0$; $\boldsymbol{x} = (1,0,6,0) + x_2(-3,1,0,0) + x_4(-2,0,-4,1)$. **Ex 1:** Condition $b_1 + b_2 + b_3 = 0$; $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{0} = (2b_1 - b_2, b_2 - b_1)$. **Ex 2:** $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & -2 & 1 \end{bmatrix}$; $\boldsymbol{x} = (2,1,0) + x_3(-3,2,1)$.

3.4 A: 1. $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 2 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 5b_1 \end{bmatrix}$. **2.** Condition: $b_3 + b_2 - 5b_1 = 0$. **3.** $C(A)$ is the plane spanned by pivot columns $(1,2,3)$ and $(3,8,7)$, with equation $b_3 + b_2 - 5b_1 = 0$. **4.** Special solutions: $\boldsymbol{s}_1 = (-2,1,0,0)$ and $\boldsymbol{s}_2 = (-2,0,-1,1)$. **5.** For $\boldsymbol{b} = (0,6,-6)$, $\boldsymbol{x}_p = (-9,0,3,0)$ and $\boldsymbol{x} = \boldsymbol{x}_p + c_1\boldsymbol{s}_1 + c_2\boldsymbol{s}_2$. **6.** $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 2 & -9 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

3.4 B: 1. Full column rank $r = n \le m$, $N(A) = \{\boldsymbol{0}\}$. **2.** $n = 2$ columns, $r = 1$, col 2 $= -$col 1, $\boldsymbol{b} =$ col 1. **3.** $r < m$, $\boldsymbol{b} \notin C(A)$, $\boldsymbol{b} \neq \boldsymbol{0}$. **4.** $n = 3$ columns, $r = 2$, $m \ge 2$, col 3 $= -$col 1, $\boldsymbol{b} =$ col 1 + col 2. **5.** $r < n$ and $\boldsymbol{b} \in C(A)$.

3.4 C: $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & -4 & 7 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, giving $\boldsymbol{x}_p = (7,0,-3,0)$, $\boldsymbol{s}_1 = (-2,1,0,0)$, $\boldsymbol{s}_2 = (4,0,-4,1)$. Row combination: $2(\text{row } 1) + 1(\text{row } 2) - 1(\text{row } 3) = \boldsymbol{0}^T$, so $\boldsymbol{y} = (2,1,-1)$ and $2(4) + 1(2) - 1(10) = 0$.

3.3 Problem Set 3.4 (Verbatim Questions and Solutions)

1. (Recommended) Execute the six steps of Worked Example **3.4 A** to describe the column space and nullspace of A and the complete solution to $A\boldsymbol{x} = \boldsymbol{b}$:

$$A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix} \qquad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}.$$

2. Carry out the same six steps for this matrix A with rank one. You will find *two* conditions on b_1, b_2, b_3 for $A\boldsymbol{x} = \boldsymbol{b}$ to be solvable. Together these two conditions put $\boldsymbol{b}$ into the _____ space (two planes give a line):

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ 4 & 2 & 6 \end{bmatrix} \qquad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \\ 20 \end{bmatrix}.$$

3. Write the complete solution as $\boldsymbol{x}_p$ plus any multiple of $\boldsymbol{s}$ in the nullspace:

$$x + 3y + 3z = 1, \qquad 2x + 6y + 9z = 5, \qquad -x - 3y + 3z = 5.$$

4. Find the complete solution (also called the *general solution*) to

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

5. Under what condition on b_1, b_2, b_3 is this system solvable? Include $\boldsymbol{b}$ as a fourth column in elimination. Find all solutions when that condition holds:

$$x + 2y - 2z = b_1, \qquad 2x + 5y - 4z = b_2, \qquad 4x + 9y - 8z = b_3.$$

6. What conditions on b_1, b_2, b_3, b_4 make each system solvable? Find $\boldsymbol{x}$ in that case:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 2 & 5 \\ 3 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}.$$

7. Show by elimination that (b_1, b_2, b_3) is in the column space if $b_3 - 2b_2 + 4b_1 = 0$.

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 8 & 2 \\ 2 & 4 & 0 \end{bmatrix}.$$ What combination of the rows of A gives the zero row?

8. Which vectors (b_1, b_2, b_3) are in the column space of A? Which combinations of the rows of A give zero?

$$\text{(a) } A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 3 \\ 0 & 2 & 5 \end{bmatrix} \qquad \text{(b) } A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}.$$

END-OF-PAGE SOLUTION Problem Set 3.4 (Problems 1–8)

1. 1. $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 & 4 & b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 2b_1 \end{bmatrix}$. **2.** $b_3 + b_2 - 2b_1 = 0$. **3.** $C(A)$ is the plane spanned by $(2,2,2)$ and $(4,5,3)$. **4.** $\boldsymbol{s}_1 = (-1,-1,1,0), \boldsymbol{s}_2 = (2,-2,0,1)$. **5.** For $\boldsymbol{b} = (4,3,5)$, $\boldsymbol{x}_p = (4,-1,0,0)$ and $\boldsymbol{x} = \boldsymbol{x}_p + c_1\boldsymbol{s}_1 + c_2\boldsymbol{s}_2$. **6.** $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & -2 & 4 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

2. 1. $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 & b_1 \\ 0 & 0 & 0 & b_2 - 3b_1 \\ 0 & 0 & 0 & b_3 - 2b_1 \end{bmatrix}$. **2.** $b_2 - 3b_1 = 0$ and $b_3 - 2b_1 = 0$ (puts $\boldsymbol{b}$ into the **column** space, line through $(1,3,2)$). **4.** $\boldsymbol{s}_1 = (-1/2,1,0), \boldsymbol{s}_2 = (-3/2,0,1)$. **5.** $\boldsymbol{x}_p = (5,0,0)$. **6.** $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix} = \begin{bmatrix} 1 & 1/2 & 3/2 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

3. $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} 1 & 3 & 3 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Longrightarrow \boldsymbol{x} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$. **4.** $\begin{bmatrix} R & \boldsymbol{d} \end{bmatrix} = \begin{bmatrix} 1 & 3 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Longrightarrow$ $\boldsymbol{x} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \\ 0 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$.

5. Condition: $b_3 - b_2 - 2b_1 = 0$. Solution: $\boldsymbol{x} = \begin{bmatrix} 5b_1 - 2b_2 \\ b_2 - 2b_1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$.

6. Left system: $b_2 = 2b_1$ and $b_4 - 3b_3 + 3b_1 = 0$; unique $\boldsymbol{x} = (5b_1 - 2b_3, b_3 - 2b_1)$. Right system: $b_2 = 2b_1$; $\boldsymbol{x} = (5b_1 - 3b_3 + b_4,\ -b_1 + 2b_3 - b_4,\ 0) + x_3(-1,-1,1)$? Wait: col 3 is $(3,6,7,12)$, and $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 0 & 0 & 0 & b_2 - 2b_1 \\ 0 & 1 & 1 & b_3 - 2b_1 \\ 0 & 3 & 3 & b_4 - 3b_1 \end{bmatrix}$, so col 3 has no pivot! Conditions are $\mathbf{b_2 - 2b_1 = 0}$ and $\mathbf{b_4 - 3b_3 + 3b_1 = 0}$, with $\boldsymbol{x} = (5b_1 - 2b_3, b_3 - 2b_1, 0) + x_3(-1,-1,1)$.

7. Elimination gives row 3 as $0 = b_3 - 2b_2 + 4b_1$. Combination: $4(\text{row } 1) - 2(\text{row } 2) + 1(\text{row } 3) = \mathbf{0}^T$.

8. (a) A is invertible ($r = 3$), so **all** $\boldsymbol{b} \in \mathbf{R}^3$ are in $C(A)$ and only $0(\text{row } 1) + 0(\text{row } 2) + 0(\text{row } 3) = \mathbf{0}^T$. **(b)** Condition $\mathbf{b_3 - 2b_2 = 0}$; row combination $0(\text{row } 1) - 2(\text{row } 2) + 1(\text{row } 3) = \mathbf{0}^T$.

9. **(a)** The Worked Example **3.4 A** reached $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix}$ from $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$. Put the multipliers into L and verify that LU equals A and $L\boldsymbol{c}$ equals $\boldsymbol{b}$.
(b) Combine the pivot columns of A with the numbers -9 and 3 in the particular solution $\boldsymbol{x}_p$. What is that linear combination and why?

10. Construct a 2 by 3 system $A\boldsymbol{x} = \boldsymbol{b}$ with particular solution $\boldsymbol{x}_p = (2,4,0)$ and homogeneous solution $\boldsymbol{x}_n =$ any multiple of $(1,1,1)$.

11. Why can't a 1 by 3 system have $\boldsymbol{x}_p = (2,4,0)$ and $\boldsymbol{x}_n =$ any multiple of $(1,1,1)$?

12. **(a)** If $A\boldsymbol{x} = \boldsymbol{b}$ has two solutions $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, find two solutions to $A\boldsymbol{x} = \boldsymbol{0}$.
(b) Then find another solution to $A\boldsymbol{x} = \boldsymbol{0}$ and another solution to $A\boldsymbol{x} = \boldsymbol{b}$.

13. Explain why these are all *false*:
(a) The complete solution is any linear combination of $\boldsymbol{x}_p$ and $\boldsymbol{x}_n$.
(b) A system $A\boldsymbol{x} = \boldsymbol{b}$ has at most one particular solution.
(c) The solution $\boldsymbol{x}_p$ with all free variables zero is the shortest solution (minimum length $\|\boldsymbol{x}\|$). Find a 2 by 2 counterexample.
(d) If A is invertible there is no solution $\boldsymbol{x}_n$ in the nullspace.

14. Suppose column 5 of U has no pivot. Then x_5 is a _____ variable. The zero vector (is) (is not) the only solution to $A\boldsymbol{x} = \boldsymbol{0}$. If $A\boldsymbol{x} = \boldsymbol{b}$ has a solution, then it has _____ solutions.

15. Suppose row 3 of U has no pivot. Then that row is _____. The equation $U\boldsymbol{x} = \boldsymbol{c}$ is only solvable provided _____. The equation $A\boldsymbol{x} = \boldsymbol{b}$ (is) (is not) (might not be) solvable.

Questions 16–20 are about matrices of "full rank" $r = m$ or $r = n$.

16. The largest possible rank of a 3 by 5 matrix is _____. Then there is a pivot in every _____ of U and R. The solution to $A\boldsymbol{x} = \boldsymbol{b}$ (*always exists*) (*is unique*). The column space of A is _____. An example is $A =$ _____.

17. The largest possible rank of a 6 by 4 matrix is _____. Then there is a pivot in every _____ of U and R. The solution to $A\boldsymbol{x} = \boldsymbol{b}$ (*always exists*) (*is unique*). The nullspace of A is _____. An example is $A =$ _____.

18. Find by elimination the rank of A and also the rank of A^T:

$$A = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 11 & 5 \\ -1 & 2 & 10 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & q \end{bmatrix} \text{ (rank depends on } q\text{)}.$$

19. Find the rank of A and also of A^TA and also of AA^T:

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

20. Reduce A to its echelon form U. Then find a triangular L so that $A = LU$.

$$A = \begin{bmatrix} 3 & 4 & 1 & 0 \\ 6 & 5 & 2 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 3 \\ 0 & 6 & 5 & 4 \end{bmatrix}.$$

21. Find the complete solution in the form $\boldsymbol{x}_p + \boldsymbol{x}_n$ to these full rank systems:
(a) $x + y + z = 4$ **(b)** $x + y + z = 4, \quad x - y + z = 4.$

22. If $A\boldsymbol{x} = \boldsymbol{b}$ has infinitely many solutions, why is it impossible for $A\boldsymbol{x} = \boldsymbol{B}$ (new right side) to have only *one* solution? Could $A\boldsymbol{x} = \boldsymbol{B}$ have *no* solution?

23. Choose the number q so that (if possible) the ranks are (a) 1, (b) 2, (c) 3:

$$A = \begin{bmatrix} 6 & 4 & 2 \\ -3 & -2 & -1 \\ 9 & 6 & q \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 1 & 3 \\ q & 2 & q \end{bmatrix}.$$

END-OF-PAGE SOLUTION Problem Set 3.4 (Problems 9–23)

9. (a) $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}$ satisfies $LU = A$ and $L\boldsymbol{c} = \boldsymbol{b}$. **(b)** $-9(1,2,3) + 3(3,8,7) = (0,6,-6) = \boldsymbol{b}$ because $A\boldsymbol{x}_p = \boldsymbol{b}$.

10. $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. **11.** A 1×3 matrix has rank ≤ 1, so it has at least $3 - 1 = 2$ free variables (cannot have a 1D nullspace).

12. (a) $\boldsymbol{0}$ and $\boldsymbol{x}_1 - \boldsymbol{x}_2$. **(b)** $2(\boldsymbol{x}_1 - \boldsymbol{x}_2)$ solves $A\boldsymbol{x} = \boldsymbol{0}$, and $\boldsymbol{x}_1 + (\boldsymbol{x}_1 - \boldsymbol{x}_2) = 2\boldsymbol{x}_1 - \boldsymbol{x}_2$ solves $A\boldsymbol{x} = \boldsymbol{b}$.

13. (a) $\boldsymbol{x}_p$ must have coefficient 1 ($A(c\boldsymbol{x}_p) = c\boldsymbol{b} \neq \boldsymbol{b}$). **(b)** Any point on the solution flat is a particular solution. **(c)** For $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\boldsymbol{x}_p = (2,0)$ has length 2, while $(1,1)$ has length $\sqrt{2} < 2$. **(d)** $\boldsymbol{x}_n = \boldsymbol{0}$ is always in $N(A)$.

14. free; **is not**; **infinitely many**. **15. all zeros**; $\mathbf{c_3 = 0}$; **might not be**.

16. 3; **row**; **always exists**; $\mathbf{R}^3$; $A = \begin{bmatrix} I_3 & 0 \end{bmatrix}$. **17. 4**; **column**; **is unique**; $\{\boldsymbol{0}\}$; $A = \begin{bmatrix} I_4 \\ 0 \end{bmatrix}$.

18. Left: $r = 2$ for both A and A^T. Right: $r = 3$ if $q \neq 2$; $r = 2$ if $q = 2$. **19.** Both matrices have $\text{rank}(A) = \text{rank}(A^TA) = \text{rank}(AA^T) = 2$.

20. Left: $U = \begin{bmatrix} 3 & 4 & 1 & 0 \\ 0 & -3 & 0 & 1 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$. Right: $U = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & -2 & 3 \\ 0 & 0 & 11 & -5 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$.

21. (a) $\boldsymbol{x} = (4,0,0) + y(-1,1,0) + z(-1,0,1)$. **(b)** $\boldsymbol{x} = (4,0,0) + z(-1,0,1)$.

22. Infinitely many solutions means $N(A)$ has nonzero vectors $\boldsymbol{x}_n$; if $A\boldsymbol{x} = \boldsymbol{B}$ has one solution $\boldsymbol{X}_p$, then $\boldsymbol{X}_p + \boldsymbol{x}_n$ gives infinitely many solutions. Yes, $A\boldsymbol{x} = \boldsymbol{B}$ can have no solution if $r < m$.

23. For A: (a) $r = 1$ if $\mathbf{q = 3}$; (b) $r = 2$ if $\mathbf{q \neq 3}$; (c) $r = 3$ is **impossible**. For B: (a) $r = 1$ if $\mathbf{q = 6}$; (b) $r = 2$ if $\mathbf{q \neq 6}$; (c) $r = 3$ is **impossible** (B has only 2 rows).

24. Give examples of matrices A for which the number of solutions to $A\boldsymbol{x} = \boldsymbol{b}$ is
(a) 0 or 1, depending on $\boldsymbol{b}$ **(b)** ∞, regardless of $\boldsymbol{b}$ **(c)** 0 or ∞, depending on $\boldsymbol{b}$ **(d)** 1, regardless of $\boldsymbol{b}$.

25. Write down all known relations between r and m and n if $A\boldsymbol{x} = \boldsymbol{b}$ has
(a) no solution for some $\boldsymbol{b}$ **(b)** infinitely many solutions for every $\boldsymbol{b}$
(c) exactly one solution for some $\boldsymbol{b}$, no solution for other $\boldsymbol{b}$ **(d)** exactly one solution for every $\boldsymbol{b}$.

Questions 26–33 are about Gauss-Jordan elimination (upwards as well as downwards) and the reduced echelon matrix R.

26. Continue elimination from U to R. Divide rows by pivots so the new pivots are all 1. Then produce zeros *above* those pivots to reach R:

$$U = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 5 \end{bmatrix}.$$

27. Suppose U is square with n pivots (an invertible matrix). *Explain why* $R = I$.

28. Apply Gauss-Jordan elimination to $U\boldsymbol{x} = \boldsymbol{0}$ and $U\boldsymbol{x} = \boldsymbol{c}$. Reach $R\boldsymbol{x} = \boldsymbol{0}$ and $R\boldsymbol{x} = \boldsymbol{d}$:

$$\begin{bmatrix} U & \boldsymbol{0} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & \boldsymbol{0} \\ 0 & 0 & 4 & \boldsymbol{0} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} U & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & \boldsymbol{5} \\ 0 & 0 & 4 & \boldsymbol{8} \end{bmatrix}.$$

Solve $R\boldsymbol{x} = \boldsymbol{0}$ to find $\boldsymbol{x}_n$ (its free variable is $x_2 = 1$). Solve $R\boldsymbol{x} = \boldsymbol{d}$ to find $\boldsymbol{x}_p$ (its free variable is $x_2 = 0$).

29. Apply Gauss-Jordan elimination to reduce to $R\boldsymbol{x} = \boldsymbol{0}$ and $R\boldsymbol{x} = \boldsymbol{d}$:

$$\begin{bmatrix} U & \boldsymbol{0} \end{bmatrix} = \begin{bmatrix} 3 & 0 & 6 & \boldsymbol{0} \\ 0 & 0 & 2 & \boldsymbol{0} \\ 0 & 0 & 0 & \boldsymbol{0} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} U & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} 3 & 0 & 6 & \boldsymbol{9} \\ 0 & 0 & 2 & \boldsymbol{4} \\ 0 & 0 & 0 & \boldsymbol{5} \end{bmatrix}.$$

Solve $U\boldsymbol{x} = \boldsymbol{0}$ or $R\boldsymbol{x} = \boldsymbol{0}$ to find $\boldsymbol{x}_n$ (free variable $= 1$). What are the solutions to $R\boldsymbol{x} = \boldsymbol{d}$?

30. Reduce to $U\boldsymbol{x} = \boldsymbol{c}$ (Gaussian elimination) and then $R\boldsymbol{x} = \boldsymbol{d}$ (Gauss-Jordan):

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix} = \boldsymbol{b}. \quad \text{Find } \boldsymbol{x}_p \text{ and all } \boldsymbol{x}_n.$$

31. Find matrices A and B with the given property or explain why you can't:
(a) The only solution of $A\boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is $\boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. **(b)** The only solution of $B\boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is $\boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

32. Find the LU factorization of A and the complete solution to $A\boldsymbol{x} = \boldsymbol{b}$:

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 5 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 5 \end{bmatrix} \quad \text{and then} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

33. The complete solution to $A\boldsymbol{x} = \begin{bmatrix}1\\3\end{bmatrix}$ is $\boldsymbol{x} = \begin{bmatrix}1\\0\end{bmatrix} + c\begin{bmatrix}0\\1\end{bmatrix}$. Find A.

34. Challenge Problems: Suppose you know that the 3 by 4 matrix A has the vector $\boldsymbol{s} = (2,3,1,0)$ as the only special solution to $A\boldsymbol{x} = \boldsymbol{0}$.
(a) What is the *rank* of A and the complete solution to $A\boldsymbol{x} = \boldsymbol{0}$?
(b) What is the exact row reduced echelon form R of A?
(c) How do you know that $A\boldsymbol{x} = \boldsymbol{b}$ can be solved for *all* $\boldsymbol{b}$?

35. Suppose K is the 9 by 9 second difference matrix (2's on the diagonal, -1's on the diagonal above and also below). Solve the equation $K\boldsymbol{x} = \boldsymbol{b} = (10,\dots,10)$. If you graph $x_1,\dots,x_9$ above the points $1,\dots,9$ on the x axis, I think the nine points fall on a parabola.

36. Suppose $A\boldsymbol{x} = \boldsymbol{b}$ and $C\boldsymbol{x} = \boldsymbol{b}$ have the same (complete) solutions for every $\boldsymbol{b}$. Is it true that $A = C$?

END-OF-PAGE SOLUTION Problem Set 3.4 (Problems 24–36)

24. (a) $A = \begin{bmatrix}1\\1\end{bmatrix}$. **(b)** $A = \begin{bmatrix}1 & 1\end{bmatrix}$. **(c)** $A = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$. **(d)** $A = I_2$.

25. (a) $r < m$. **(b)** $r = m < n$. **(c)** $r = n < m$. **(d)** $r = m = n$.

26. Left: $R = \begin{bmatrix}1 & 0 & -2\\0 & 1 & 2\\0 & 0 & 0\end{bmatrix}$. Right: $R = I_3$. **27.** Every column is a pivot column, so R has 1s on the diagonal and 0s below and above ($R = I_n$).

28. $\begin{bmatrix}R & \boldsymbol{0}\end{bmatrix} = \begin{bmatrix}1 & 2 & 0 & 0\\0 & 0 & 1 & 0\end{bmatrix} \implies \boldsymbol{x}_n = (-2,1,0)$. $\begin{bmatrix}R & \boldsymbol{d}\end{bmatrix} = \begin{bmatrix}1 & 2 & 0 & -1\\0 & 0 & 1 & 2\end{bmatrix} \implies \boldsymbol{x}_p = (-1,0,2)$.

29. $\begin{bmatrix}R & \boldsymbol{0}\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & 0\end{bmatrix} \implies \boldsymbol{x}_n = (0,1,0)$. For $\begin{bmatrix}U & \boldsymbol{c}\end{bmatrix}$, row 3 is $0 = 5$, so **no solution**.

30. $\begin{bmatrix}U & \boldsymbol{c}\end{bmatrix} = \begin{bmatrix}1 & 0 & 2 & 3 & 2\\0 & 3 & 0 & -3 & 3\\0 & 0 & 0 & 3 & 6\end{bmatrix} \to \begin{bmatrix}R & \boldsymbol{d}\end{bmatrix} = \begin{bmatrix}1 & 0 & 2 & 0 & -4\\0 & 1 & 0 & 0 & 3\\0 & 0 & 0 & 1 & 2\end{bmatrix}$. $\boldsymbol{x}_p = (-4,3,0,2)$, $\boldsymbol{x}_n = x_3(-2,0,1,0)$.

31. (a) $A = \begin{bmatrix}1 & 1\\0 & 2\\0 & 3\end{bmatrix}$. **(b) Impossible**: B is 2×3 ($n > m$), so $N(B)$ has at least $3 - 2 = 1$ free variable (cannot have a unique solution).

32. $L = \begin{bmatrix}1 & 0 & 0 & 0\\1 & 1 & 0 & 0\\2 & 2 & 1 & 0\\1 & 2 & 0 & 1\end{bmatrix}, U = \begin{bmatrix}1 & 3 & 1\\0 & -1 & 2\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix}$. For $\boldsymbol{b} = (1,3,6,5)$: $\boldsymbol{x} = (-5,2,0) + x_3(-7,2,1)$. For $\boldsymbol{b} = (1,0,0,0)$: $L\boldsymbol{c} = \boldsymbol{b} \implies \boldsymbol{c} = (1,-1,0,1)$, so row 4 gives $0 = 1$ (**no solution**).

33. $A = \begin{bmatrix}1 & 0\\3 & 0\end{bmatrix}$. **34. (a)** $r = 4 - 1 = 3$; $\boldsymbol{x} = c(2,3,1,0)$. **(b)** Free variable is x_3, pivot cols 1,2,4: $R = \begin{bmatrix}1 & 0 & -2 & 0\\0 & 1 & -3 & 0\\0 & 0 & 0 & 1\end{bmatrix}$. **(c)** $r = m = 3$ (full row rank).

35. $x_k = 5k(10-k)$ for $k = 1,\dots,9$: $(45,80,105,120,125,120,105,80,45)$, which lies on the parabola $5x(10-x)$.

36. Yes: choosing $\boldsymbol{b} = A\boldsymbol{e}_j$ shows $\boldsymbol{x} = \boldsymbol{e}_j$ solves $A\boldsymbol{x} = \boldsymbol{b}$, so $\boldsymbol{e}_j$ must also solve $C\boldsymbol{x} = \boldsymbol{b}$, meaning $C\boldsymbol{e}_j = \boldsymbol{b} = A\boldsymbol{e}_j$ for every column j.

3.4 Independence, Basis, and Dimension (Section 3.5)

GOAL

Derive the tests for linear independence, spanning sets, basis, uniqueness of basis representation, and the theorem that every basis for a vector space has the same number of vectors (the dimension).

Step 1: Define Linear Independence. The columns of A are **linearly independent** when the only solution to $A\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$ ($N(A) = \{\mathbf{0}\}$, rank $r = n$). For any sequence of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ in any vector space:

$$\boxed{x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_n\boldsymbol{v}_n = \mathbf{0} \quad \text{only happens when all } x_i = 0.} \tag{3.3}$$

If $n > m$, any n vectors in $\mathbf{R}^m$ are **automatically linearly dependent** because $A\boldsymbol{x} = \mathbf{0}$ has at least $n - m \geq 1$ free variables.

Step 2: Define Spanning Sets, Row Space, and Basis.

- A set of vectors **spans** a space if their linear combinations fill the space.
- The **row space** of A is the subspace of $\mathbf{R}^n$ spanned by the rows of A. Written as column vectors, the row space of A is $C(A^T)$.
- A **basis** for a vector space is a sequence of vectors that is **(1) linearly independent** and **(2) spans the space**.

Step 3: Prove Uniqueness of Basis Representation and Invariance of Dimension. **Uniqueness:** If $\boldsymbol{v} = a_1\boldsymbol{v}_1 + \cdots + a_n\boldsymbol{v}_n = b_1\boldsymbol{v}_1 + \cdots + b_n\boldsymbol{v}_n$, subtract to get $(a_1 - b_1)\boldsymbol{v}_1 + \cdots + (a_n - b_n)\boldsymbol{v}_n = \mathbf{0}$. By linear independence (3.3), $a_i - b_i = 0 \implies a_i = b_i$ for every i.

Invariance of Dimension ($m = n$): Suppose $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ and $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$ are both bases for the same space, and suppose $n > m$. Express each $\boldsymbol{w}_j$ as a combination of the $\boldsymbol{v}$'s: $W = VA$ where A is an $m \times n$ short wide matrix ($n > m$). Then $A\boldsymbol{x} = \mathbf{0}$ has a nonzero solution $\boldsymbol{x} \neq \mathbf{0}$, which gives $W\boldsymbol{x} = VA\boldsymbol{x} = \mathbf{0}$—contradicting the independence of the $\boldsymbol{w}$'s! Similarly $m > n$ is impossible. Thus $m = n$.

$$\boxed{\text{The } \textbf{dimension} \text{ of a space is the number of vectors in every basis.}} \tag{3.4}$$

For any $m \times n$ matrix A of rank r: the r **pivot columns of** A form a basis for $C(A)$ ($\dim C(A) = r$), and the r **nonzero rows of** R form a basis for the row space $C(A^T)$ ($\dim C(A^T) = r$).

3.5 In-Text and Worked Examples (Section 3.5)

BOOK EXAMPLE 3.5 In-Text 1–9 (p. 170–176) Independence, Row Space, and Basis

Ex 1: Show that the columns of $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \\ 1 & 0 & 3 \end{bmatrix}$ are dependent by solving $A\boldsymbol{x} = \mathbf{0}$.

Ex 2–4: Check spanning and independence in $\mathbf{R}^2$ for $\{(1,0),(0,1)\}$, $\{(1,0),(0,1),(4,7)\}$, and $\{(1,1),(-1,-1)\}$.

Ex 5: Describe the column space and row space of $A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \\ 3 & 5 \end{bmatrix}$.

Ex 6–9: Give bases for $C(A)$ and $C(A^T)$ for $A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$ and $R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

BOOK EXAMPLE 3.5 A Span, Dimension, Column Space, Nullspace, and Basis Extension

Start with the vectors $v_1 = (1,2,0)$ and $v_2 = (2,3,0)$. **(a)** Are they linearly independent? **(b)** Are they a basis for any space? **(c)** What space **V** do they span? **(d)** What is the dimension of **V**? **(e)** Which matrices A have **V** as their column space? **(f)** Which matrices have **V** as their nullspace? **(g)** Describe all vectors v_3 that complete a basis v_1, v_2, v_3 for $\mathbf{R}^3$.

BOOK EXAMPLE 3.5 B Testing Independence after a Change of Basis $V = WM$

Start with three independent vectors w_1, w_2, w_3. Take combinations of those vectors to produce v_1, v_2, v_3. Write the combinations in matrix form as $V = WM$:

$$\begin{aligned} v_1 &= w_1 + w_2 \\ v_2 &= w_1 + 2w_2 + w_3 \\ v_3 &= \quad w_2 + cw_3 \end{aligned} \quad \text{which is} \quad \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & c \end{bmatrix}.$$

What is the test on a matrix V to see if its columns are linearly independent? If $c \neq 1$ show that v_1, v_2, v_3 are linearly independent. If $c = 1$ show that the v's are linearly *dependent*.

BOOK EXAMPLE 3.5 C Invertible Matrix Transformation of a Basis

(**Important example**) Suppose $v_1, \ldots, v_n$ is a basis for $\mathbf{R}^n$ and the n by n matrix A is invertible. Show that $Av_1, \ldots, Av_n$ is also a basis for $\mathbf{R}^n$.

END-OF-PAGE SOLUTION Section 3.5 In-Text 1–9 & Worked Examples 3.5 A, 3.5 B, 3.5 C

In-Text 1–9: Ex 1: $A \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} = \mathbf{0}$, so columns are dependent ($r = 2$). **Ex 2–4:** $\{(1,0),(0,1)\}$ is a basis for $\mathbf{R}^2$; $\{(1,0),(0,1),(4,7)\}$ spans $\mathbf{R}^2$ but is dependent; $\{(1,1),(-1,-1)\}$ spans only a line and is dependent. **Ex 5:** $C(A)$ is a plane in $\mathbf{R}^3$; $C(A^T)$ is all of $\mathbf{R}^2$. **Ex 8:** Basis for $C(A)$ is $\{(2,3)\}$; basis for $C(A^T)$ is $\{(1,2)\}$. **Ex 9:** Pivot columns $(1,0,0)$ and $(0,1,0)$ form a basis for $C(R)$ (all pairs except columns 1 and 2 work); nonzero rows $(1,2,0,3)$ and $(0,0,1,4)$ form a basis for $C(R^T)$.

3.5 A: (a) Yes, independent. **(b)** Yes, basis for their span **V**. **(c)** **V** is the xy plane $\{(x,y,0)\}$ in $\mathbf{R}^3$. **(d)** $\dim(\mathbf{V}) = 2$. **(e)** Any $3 \times n$ matrix A of rank 2 whose columns are combinations of v_1, v_2 (e.g. $A = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$). **(f)** Any $m \times 3$ matrix B of rank 1 whose rows are multiples of $(0,0,1)$. **(g)** Any vector $v_3 = (a,b,c)$ with $c \neq 0$.

3.5 B: Test: $N(V) = \{\mathbf{0}\}$. If $c = 1$, $\boldsymbol{x} = (1,-1,1) \in N(M)$, so $V\boldsymbol{x} = WM\boldsymbol{x} = \mathbf{0}$ ($v_1 - v_2 + v_3 = \mathbf{0}$, dependent). If $c \neq 1$, M is invertible, so $V\boldsymbol{x} = \mathbf{0} \implies WM\boldsymbol{x} = \mathbf{0} \implies M\boldsymbol{x} = \mathbf{0}$ (by independence of w's) $\implies \boldsymbol{x} = M^{-1}\mathbf{0} = \mathbf{0}$ (independent).

3.5 C: If $\sum c_i A v_i = A(\sum c_i v_i) = \mathbf{0}$, multiply by A^{-1} to get $\sum c_i v_i = \mathbf{0} \implies$ all $c_i =$

0 (independent). Since n independent vectors in $\mathbf{R}^n$ automatically span $\mathbf{R}^n$ (or solve $\sum c_i A\boldsymbol{v}_i = \boldsymbol{b} \iff \sum c_i \boldsymbol{v}_i = A^{-1}\boldsymbol{b}$), $A\boldsymbol{v}_1, \ldots, A\boldsymbol{v}_n$ is a basis for $\mathbf{R}^n$.

3.6 Problem Set 3.5 (Verbatim Questions and Solutions)

Questions 1–10 are about linear independence and linear dependence.

1. Show that v_1, v_2, v_3 are independent but v_1, v_2, v_3, v_4 are dependent:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \qquad v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad v_4 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

2. (Recommended) Find the largest possible number of independent vectors among

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad v_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad v_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \quad v_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

3. Prove that if $a = 0$ or $d = 0$ or $f = 0$ (3 cases), the columns of U are dependent:

$$U = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}.$$

4. If a, d, f in Question 3 are all nonzero, show that the only solution to $Ux = 0$ is $x = 0$. Then the upper triangular U has independent columns.

5. Decide the dependence or independence of
(a) the vectors $(1,3,2)$ and $(2,1,3)$ and $(3,2,1)$ **(b)** the vectors $(1,-3,2)$ and $(2,1,-3)$ and $(-3,2,1)$.

6. Choose three independent columns of U. Then make two other choices. Do the same for A.

$$U = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 4 & 6 & 8 & 2 \end{bmatrix}.$$

7. If w_1, w_2, w_3 are independent vectors, show that the differences $v_1 = w_2 - w_3$ and $v_2 = w_1 - w_3$ and $v_3 = w_1 - w_2$ are *dependent*. Find a combination of the v's that gives zero. Which matrix A in $\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} A$ is singular?

8. If w_1, w_2, w_3 are independent vectors, show that the sums $v_1 = w_2 + w_3$ and $v_2 = w_1 + w_3$ and $v_3 = w_1 + w_2$ are *independent*. (Write $c_1v_1 + c_2v_2 + c_3v_3 = 0$ in terms of the w's. Find and solve equations for the c's, to show they are zero.)

9. Suppose v_1, v_2, v_3, v_4 are vectors in $\mathbf{R}^3$.
(a) These four vectors are dependent because _____.
(b) The two vectors v_1 and v_2 will be dependent if _____.
(c) The vectors v_1 and $(0,0,0)$ are dependent because _____.

10. Find two independent vectors on the plane $x + 2y - 3z - t = 0$ in $\mathbf{R}^4$. Then find three independent vectors. Why not four? This plane is the nullspace of what matrix?

Questions 11–15 are about the space *spanned* by a set of vectors.

11. Describe the subspace of $\mathbf{R}^3$ (is it a line or plane or $\mathbf{R}^3$?) spanned by
(a) the two vectors $(1,1,-1)$ and $(-1,-1,1)$ **(b)** the three vectors $(0,1,1)$ and $(1,1,0)$ and

$(0,0,0)$

(c) all vectors in $\mathbf{R}^3$ with whole number components **(d)** all vectors with positive components.

12. The vector $\boldsymbol{b}$ is in the subspace spanned by the columns of A when _____ has a solution. The vector $\boldsymbol{c}$ is in the row space of A when _____ has a solution.

True or false: If the zero vector is in the row space, the rows are dependent.

13. Find the dimensions of these 4 spaces. Which two of the spaces are the same? (a) column space of A, (b) column space of U, (c) row space of A, (d) row space of U:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

14. $v + w$ and $v - w$ are combinations of v and w. Write v and w as combinations of $v + w$ and $v - w$. The two pairs of vectors _____ the same space. When are they a basis for the same space?

END-OF-PAGE SOLUTION Problem Set 3.5 (Problems 1–14)

1. $\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$ is upper triangular with 1s on the diagonal (invertible, independent). Four vectors in $\mathbf{R}^3$ are always dependent: $v_1 + v_2 - 4v_3 + v_4 = \mathbf{0}$.

2. **3** independent vectors (e.g. v_1, v_2, v_3). Every v_i has components summing to 0, so all 6 lie in the 3D hyperplane $x_1 + x_2 + x_3 + x_4 = 0$.

3. If $a = 0$, $\boldsymbol{x} = (1,0,0)$ solves $U\boldsymbol{x} = \mathbf{0}$. If $d = 0$, $\boldsymbol{x} = (-b, a, 0)$ solves $U\boldsymbol{x} = \mathbf{0}$. If $f = 0$, there are at most 2 pivots in 3 columns.

4. Back substitution: $fx_3 = 0 \implies x_3 = 0$; $dx_2 = 0 \implies x_2 = 0$; $ax_1 = 0 \implies x_1 = 0$.

5. (a) Independent (row reduction gives 3 pivots: $1, -5, -18/5$). **(b) Dependent** ($v_1 + v_2 + v_3 = \mathbf{0}$).

6. Columns $(1,2,4)$ (pivot columns), or $(1,3,4)$, or $(2,3,4)$ for both U and A (since row 4 of A is $2 \times$ row 1).

7. $v_1 - v_2 + v_3 = \mathbf{0}$. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$ is singular. **8.** $(c_2 + c_3)\boldsymbol{w}_1 + (c_1 + c_3)\boldsymbol{w}_2 + (c_1 + c_2)\boldsymbol{w}_3 = \mathbf{0} \implies c_2 + c_3 = c_1 + c_3 = c_1 + c_2 = 0 \implies c_1 = c_2 = c_3 = 0$.

9. (a) $n = 4 > m = 3$. **(b)** one is a scalar multiple of the other. **(c)** $0v_1 + 1(0,0,0) = (0,0,0)$ has a nonzero coefficient.

10. Special solutions $(-2,1,0,0)$, $(3,0,1,0)$, $(1,0,0,1)$ are 3 independent vectors. Not four because the hyperplane is 3-dimensional ($N(\begin{bmatrix} 1 & 2 & -3 & -1 \end{bmatrix})$).

11. (a) Line. **(b) Plane**. **(c) All of $\mathbf{R}^3$**. **(d) All of $\mathbf{R}^3$**.

12. $\mathbf{Ax} = \mathbf{b}$; $\mathbf{A^T y} = \mathbf{c}$. **False**: the zero vector is in *every* subspace, even when the rows are independent.

13. All four spaces have dimension **2**. The **row space of A (c)** and the **row space of U (d)** are the same.

14. $v = \frac{1}{2}(v + w) + \frac{1}{2}(v - w)$ and $w = \frac{1}{2}(v + w) - \frac{1}{2}(v - w)$. They **span** the same space. They are a basis when v and w are **linearly independent**.

Questions 15–25 are about the requirements for a basis.

15. If $v_1, \ldots, v_n$ are linearly independent, the space they span has dimension _____. These vectors are a _____ for that space. If the vectors are the columns of an m by n matrix, then m is _____ than n. If $m = n$, that matrix is _____.

16. Find a basis for each of these subspaces of $\mathbf{R}^4$:

(a) All vectors whose components are equal. **(b)** All vectors whose components add to zero.

(c) All vectors that are perpendicular to $(1,1,0,0)$ and $(1,0,1,1)$. **(d)** The column space and the nullspace of I (4 by 4).

17. Find three different bases for the column space of $U = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$. Then find two different bases for the row space of U.

18. Suppose $v_1, v_2, \ldots, v_6$ are six vectors in $\mathbf{R}^4$.

(a) Those vectors (do)(do not)(might not) span $\mathbf{R}^4$. **(b)** Those vectors (are)(are not)(might be) linearly independent.

(c) Any four of those vectors (are)(are not)(might be) a basis for $\mathbf{R}^4$.

19. The columns of A are n vectors from $\mathbf{R}^m$. If they are linearly independent, what is the rank of A? If they span $\mathbf{R}^m$, what is the rank? If they are a basis for $\mathbf{R}^m$, what then? *Looking ahead:* The rank r counts the number of _____ columns.

20. Find a basis for the plane $x - 2y + 3z = 0$ in $\mathbf{R}^3$. Then find a basis for the intersection of that plane with the xy plane. Then find a basis for all vectors perpendicular to the plane.

21. Suppose the columns of a 5 by 5 matrix A are a basis for $\mathbf{R}^5$.

(a) The equation $A\boldsymbol{x} = \mathbf{0}$ has only the solution $\boldsymbol{x} = \mathbf{0}$ because _____.

(b) If $\boldsymbol{b}$ is in $\mathbf{R}^5$ then $A\boldsymbol{x} = \boldsymbol{b}$ is solvable because the basis vectors _____ $\mathbf{R}^5$.

22. Suppose $\mathbf{S}$ is a 5-dimensional subspace of $\mathbf{R}^6$. True or false (example if false):

(a) Every basis for $\mathbf{S}$ can be extended to a basis for $\mathbf{R}^6$ by adding one more vector.

(b) Every basis for $\mathbf{R}^6$ can be reduced to a basis for $\mathbf{S}$ by removing one vector.

23. U comes from A by subtracting row 1 from row 3:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Find bases for the two column spaces. Find bases for the two row spaces. Find bases for the two nullspaces. Which spaces stay fixed in elimination?

24. True or false (give a good reason):

(a) If the columns of a matrix are dependent, so are the rows.

(b) The column space of a 2 by 2 matrix is the same as its row space.

(c) The column space of a 2 by 2 matrix has the same dimension as its row space.

(d) The columns of a matrix are a basis for the column space.

25. For which numbers c and d do these matrices have rank 2?

$$A = \begin{bmatrix} 1 & 2 & 5 & 0 & 5 \\ 0 & 0 & c & 2 & 2 \\ 0 & 0 & 0 & d & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} c & d \\ d & c \end{bmatrix}.$$

END-OF-PAGE SOLUTION Problem Set 3.5 (Problems 15–25)

15. **n**; **basis**; **greater than or equal to** $(m \geq n)$; **invertible**.
16. (a) $\{(1,1,1,1)\}$. **(b)** $\{(-1,1,0,0),(-1,0,1,0),(-1,0,0,1)\}$. **(c)** $\{(-1,1,1,0),(-1,1,0,1)\}$. **(d)** $C(I)$: standard basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4$; $N(I)$: empty basis $\emptyset$.
17. $C(U)$: cols $(1,2)$, $(1,4)$, or $(3,2)$. Row space: rows 1 and 2 of U, or row 1 and (row 1 + row 2).
18. (a) might not. **(b) are not** $(6 > 4)$. **(c) might be**. **19.** $r = n$; $r = m$; $r = m = n$; **independent** (pivot) columns.
20. Plane basis: $\{(2,1,0),(-3,0,1)\}$. Intersection with xy plane $(z = 0)$: $\{(2,1,0)\}$. Perpendicular line: $\{(1,-2,3)\}$.
21. (a) the columns are linearly independent. **(b)** span.
22. (a) True (add any vector in $\mathbf{R}^6$ not in **S**). **(b) False**: if **S** is the hyperplane $x_1 + x_2 + \cdots + x_6 = 0$, none of the standard basis vectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_6$ lie in **S**!
23. $C(A)$: $\{(1,0,1),(3,1,3)\}$; $C(U)$: $\{(1,0,0),(3,1,0)\}$. Row spaces (both): $\{(1,3,2),(0,1,1)\}$. Nullspaces (both): $\{(1,-1,1)\}$. The **row space** and **nullspace** stay fixed.
24. (a) False (a 2×3 matrix with rank 2 has dependent columns but independent rows). **(b) False** ($\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$). **(c) True** (dim $= r$ for both). **(d) False** (only the *pivot* columns form a basis when columns are dependent).
25. For A: $\mathbf{c} = \mathbf{0}$ and $\mathbf{d} = \mathbf{0}$ (so row 3 has pivot in col 5, giving 2 pivots total). For B: $\mathbf{c} \neq \pm\mathbf{d}$.

Questions 26–30 are about spaces where the "vectors" are matrices.

26. Find a basis (and the dimension) for each of these subspaces of 3 by 3 matrices:
(a) All diagonal matrices. **(b)** All symmetric matrices ($A^T = A$). **(c)** All skew-symmetric matrices ($A^T = -A$).

27. Construct six linearly independent 3 by 3 echelon matrices $U_1, \ldots, U_6$.

28. Find a basis for the space of all 2 by 3 matrices whose columns add to zero. Find a basis for the subspace whose rows also add to zero.

29. What subspace of 3 by 3 matrices is spanned (take all combinations) by
(a) the invertible matrices? **(b)** the rank one matrices? **(c)** the identity matrix?

30. Find a basis for the space of 2 by 3 matrices whose nullspace contains $(2, 1, 1)$.

Questions 31–35 are about spaces where the "vectors" are functions.

31. **(a)** Find all functions that satisfy $\frac{dy}{dx} = 0$. **(b)** Choose a particular function that satisfies $\frac{dy}{dx} = 3$. **(c)** Find all functions that satisfies $\frac{dy}{dx} = 3$.

32. The cosine space $\mathbf{F}_3$ contains all combinations $y(x) = A\cos x + B\cos 2x + C\cos 3x$. Find a basis for the subspace with $y(0) = 0$.

33. Find a basis for the space of functions that satisfy **(a)** $\frac{dy}{dx} - 2y = 0$ **(b)** $\frac{dy}{dx} - \frac{y}{x} = 0$.

34. Suppose $y_1(x), y_2(x), y_3(x)$ are three different functions of x. The vector space they span could have dimension 1, 2, or 3. Give an example of y_1, y_2, y_3 to show each possibility.

35. Find a basis for the space of polynomials $p(x)$ of degree ≤ 3. Find a basis for the subspace with $p(1) = 0$.

36. Find a basis for the space $\mathbf{S}$ of vectors (a, b, c, d) with $a + c + d = 0$ and also for the space $\mathbf{T}$ with $a + b = 0$ and $c = 2d$. What is the dimension of the intersection $\mathbf{S} \cap \mathbf{T}$?

37. If $AS = SA$ for the *shift matrix* $S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, show that A must have the special form $A = \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}$. "The subspace of matrices that commute with the shift S has dimension _____."

38. Which of the following are bases for $\mathbf{R}^3$?
(a) $(1, 2, 0)$ and $(0, 1, -1)$ **(b)** $(1, 1, -1), (2, 3, 4), (4, 1, -1), (0, 1, -1)$
(c) $(1, 2, 2), (-1, 2, 1), (0, 8, 0)$ **(d)** $(1, 2, 2), (-1, 2, 1), (0, 8, 6)$.

39. Suppose A is 5 by 4 with rank 4. Show that $A\boldsymbol{x} = \boldsymbol{b}$ has no solution when the 5 by 5 matrix $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ is invertible. Show that $A\boldsymbol{x} = \boldsymbol{b}$ is solvable when $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ is singular.

40. **(a)** Find a basis for all solutions to $d^4y/dx^4 = y(x)$. **(b)** Find a particular solution to $d^4y/dx^4 = y(x) + 1$. Find the complete solution.

41. **Challenge Problems:** Write the 3 by 3 identity matrix as a combination of the other five permutation matrices! Then show that those five matrices are linearly independent.

42. Choose $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ in $\mathbf{R}^4$. It has 24 rearrangements. Find specific vectors $\boldsymbol{x}$ so that the dimension of their span $\mathbf{S}$ is: (a) zero, (b) one, (c) three, (d) four.

43. Intersections and sums have $\dim(\mathbf{V}) + \dim(\mathbf{W}) = \dim(\mathbf{V} \cap \mathbf{W}) + \dim(\mathbf{V} + \mathbf{W})$. Prove that the $\boldsymbol{u}$'s, $\boldsymbol{v}$'s, and $\boldsymbol{w}$'s together are independent.

44. For an m by n matrix of rank r, what are the 3 dimensions in dimension of outputs + dimension of nullspace = dimension of inputs?

45. Inside $\mathbf{R}^n$, suppose $\dim(\mathbf{V}) + \dim(\mathbf{W}) > n$. Show that some nonzero vector is in both $\mathbf{V}$ and $\mathbf{W}$.

46. Suppose A is 10 by 10 and $A^2 = 0$ (zero matrix). This means that the column space of A is contained in the _____. If A has rank r, those subspaces have dimension $r \leq 10 - r$. So the rank is $r \leq 5$.

END-OF-PAGE SOLUTION Problem Set 3.5 (Problems 26–46)

26. (a) Dim 3: E_{11}, E_{22}, E_{33}. **(b)** Dim 6: $E_{11}, E_{22}, E_{33}, E_{12} + E_{21}, E_{13} + E_{31}, E_{23} + E_{32}$. **(c)** Dim 3: $E_{12} - E_{21}, E_{13} - E_{31}, E_{23} - E_{32}$.

27. The 6 upper-triangular matrices E_{ij} $(1 \leq i \leq j \leq 3)$ (replace zero pivot above by 1 on the diagonal if strict staircase required: $E_{11}, E_{11} + E_{12}, E_{11} + E_{13}, E_{11} + E_{22}, E_{11} + E_{22} + E_{23}, E_{11} + E_{22} + E_{33}$).

28. Columns add to **0** (dim 4): $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$. Rows AND columns add to 0 (dim 2): $\begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$.

29. (a) All 3×3 matrices. **(b)** All 3×3 matrices. **(c)** Multiples cI.

30. Dim 4: $\begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & -2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \end{bmatrix}$.

31. (a) $y(x) = C$. **(b)** $y_p(x) = 3x$. **(c)** $y(x) = 3x + C$. **32.** $A + B + C = 0$: basis $\{\cos x - \cos 2x,\ \cos x - \cos 3x\}$.

33. (a) $\{e^{2x}\}$. **(b)** $\{x\}$. **34.** Dim 1: $x, 2x, 3x$; Dim 2: $1, x, 1 + x$; Dim 3: $1, x, x^2$.

35. $\mathbf{P}_3$ basis: $\{1, x, x^2, x^3\}$. With $p(1) = 0$: $\{x - 1, x^2 - 1, x^3 - 1\}$.

36. **S** basis (dim 3): $\{(0,1,0,0), (-1,0,1,0), (-1,0,0,1)\}$. **T** basis (dim 2): $\{(1,-1,0,0), (0,0,2,1)\}$. $\mathbf{S} \cap \mathbf{T}$ has dim **1** (spanned by $(-3,3,2,1)$). **37.** Dimension **3** (basis I, S, S^2).

38. Only **(c)** is a basis for $\mathbf{R}^3$ (in (d), $2(1,2,2) + 2(-1,2,1) = (0,8,6)$ is dependent).

39. $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ invertible $\implies$ $\boldsymbol{b}$ is independent of the columns of A $\implies$ $\boldsymbol{b} \notin C(A)$. Singular $\implies$ $\boldsymbol{b}$ is a combination of the 4 independent columns of A.

40. (a) $\{e^x, e^{-x}, \cos x, \sin x\}$. **(b)** $y_p(x) = -1$; $y(x) = -1 + c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$.

41. $I = \frac{1}{2}(P_{231} + P_{312}) - \frac{1}{2}$? Wait: sum of the 2 cyclic permutations $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, and sum of the 3 transpositions $P_{213} + P_{132} + P_{321} = I + \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Therefore $\mathbf{I} = \mathbf{P_{213}} + \mathbf{P_{132}} + \mathbf{P_{321}} - \mathbf{P_{231}} - \mathbf{P_{312}}$.

42. (a) $(0,0,0,0)$; (b) $(1,1,1,1)$; (c) $(1,1,-1,-1)$ (sum of components is 0); (d) $(1,0,0,0)$.

43–45. In #44: outputs $= r$, nullspace $= n - r$, inputs $= n$. In #45: $\dim(\mathbf{V} \cap \mathbf{W}) = \dim(\mathbf{V}) + \dim(\mathbf{W}) - \dim(\mathbf{V} + \mathbf{W}) > n - n = 0 \implies \dim(\mathbf{V} \cap \mathbf{W}) \geq 1$. **46.** Contained in the **nullspace** of A.

3.7 Dimensions of the Four Subspaces (Section 3.6)

GOAL

Define the Four Fundamental Subspaces of an $m \times n$ matrix A, construct a basis for each subspace from $EA = R$, and prove Part 1 of the Fundamental Theorem of Linear Algebra.

Step 1: Define the Four Fundamental Subspaces.

1. **Row space** $C(A^T)$**:** All combinations of the rows of A (a subspace of $\mathbf{R}^n$).
2. **Nullspace** $N(A)$**:** All solutions $\boldsymbol{x}$ to $A\boldsymbol{x} = \mathbf{0}$ (a subspace of $\mathbf{R}^n$).
3. **Column space** $C(A)$**:** All combinations of the columns of A (a subspace of $\mathbf{R}^m$).
4. **Left nullspace** $N(A^T)$**:** All solutions $\boldsymbol{y}$ to $A^T\boldsymbol{y} = \mathbf{0}$, or $\boldsymbol{y}^T A = \mathbf{0}^T$ (a subspace of $\mathbf{R}^m$).

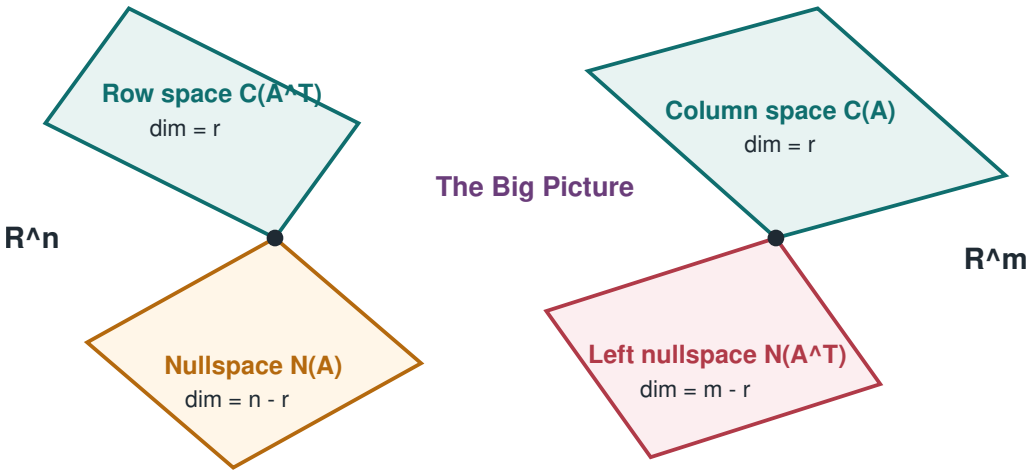


Fig. 3.2 The Big Picture: Dimensions of the Four Fundamental Subspaces in $\mathbf{R}^n$ and $\mathbf{R}^m$.

Step 2: Prove Part 1 of the Fundamental Theorem of Linear Algebra from $EA = R$. Let E be the invertible $m \times m$ product of elementary matrices that reduces A to $R = \text{rref}(A)$ with r pivots:

RESULT (Fundamental Theorem of Linear Algebra, Part 1)

Subspace	Ambient Space	Dimension	How to Construct a Basis from $EA = R$
1. Row space $C(A^T)$	$\mathbf{R}^n$	r	The first r nonzero rows of R ($C(A^T) = C(R^T)$)
2. Nullspace $N(A)$	$\mathbf{R}^n$	$n - r$	The $n - r$ special solutions to $A\boldsymbol{x} = R\boldsymbol{x} = \mathbf{0}$
3. Column space $C(A)$	$\mathbf{R}^m$	r	The r pivot columns of A itself ($C(A) \neq C(R)$)
4. Left nullspace $N(A^T)$	$\mathbf{R}^m$	$m - r$	The last $m - r$ rows of E (which produce zero rows in R)

$$\dim C(A^T) + \dim N(A) = r + (n-r) = n \quad \dim C(A) + \dim N(A^T) = r + (m-r) = m \tag{3.5}$$

3.8 In-Text and Worked Examples (Section 3.6)

BOOK EXAMPLE 3.6 In-Text 0–2 (p. 184–188) Four Subspaces of R, A, and Rank-One Matrices

In-Text 0 (p. 184–186): Find bases and dimensions of all four subspaces for $R = \begin{bmatrix} 1 & 3 & 5 & 0 & 7 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 3 & 5 & 0 & 7 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & 3 & 5 & 1 & 9 \end{bmatrix}$.

Ex 1 & 2 (p. 188): Find the four subspaces for $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. If A has three equal rows, what is its rank and what are two $\boldsymbol{y}$'s in its left nullspace?

BOOK EXAMPLE 3.6 A Bases and Dimensions of the Four Subspaces from $A = LU = E^{-1}R$

Find bases and dimensions for all four fundamental subspaces if you know that

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 & 5 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} = LU = E^{-1}R.$$

By changing only *one* number in R, change the dimensions of all four subspaces.

BOOK EXAMPLE 3.6 B Extremal Ranks and Dimensions for Sparse Matrices

Put four 1's into a 5 by 6 matrix of zeros, keeping the dimension of its *row space* as small as possible. Describe all the ways to make the dimension of its *column space* as small as possible. Describe all the ways to make the dimension of its *nullspace* as small as possible. How to make the *sum of the dimensions of all four subspaces* small?

END-OF-PAGE SOLUTION Section 3.6 In-Text & Worked Examples 3.6 A, 3.6 B

In-Text 0: $r = 2, m = 3, n = 5$. Row space (dim 2): $(1,3,5,0,7), (0,0,0,1,2)$. Nullspace (dim 3): $(-3,1,0,0,0), (-5,0,1,0,0), (-7,0,0,-2,1)$. $C(R)$ (dim 2): $(1,0,0), (0,1,0)$, whereas $C(A)$ (dim 2): $(1,0,1), (0,1,1)$. $N(R^T)$ (dim 1): $(0,0,1)$, whereas $N(A^T)$ (dim 1): $(1,1,-1)$. **Ex 1 & 2:** For $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$, $C(A) = \{c(1,2)\}$ and $N(A^T) = \{c(2,-1)\}$. If A has 3 equal nonzero rows, $r = \mathbf{1}$, and two vectors in $N(A^T)$ are $(1,-1,0)$ and $(1,0,-1)$.

3.6 A: $r = 2$ (pivot columns 1 and 3). **Row space** (dim 2): $(1,3,0,5)$ and $(0,0,1,6)$ from R. **Column space** (dim 2): $(1,2,5)$ and $(0,1,0)$ (columns 1 and 2 of E^{-1}, which are columns 1 and 3 of A). **Nullspace** (dim 2): $(-3,1,0,0)$ and $(-5,0,-6,1)$ from R. **Left nullspace** $N(A^T)$ (dim 1): $(-5,0,1)$ (row 3 of $E = L^{-1}$). Changing the $(3,2)$ or $(3,4)$ entry of R from 0 to 1 increases r from 2 to 3 and changes all four dimensions.

3.6 B: To minimize $\dim C(A^T) = \dim C(A) = r = 1$, place the four 1's in the same row, in the same column, or in a 2×2 block $a_{ii} = a_{ij} = a_{ji} = a_{jj} = 1$. To minimize $\dim N(A) = 6 - r = 2$, maximize $r = 4$ by placing the four 1's in four different rows and four different columns. The sum of all four dimensions is always $r + (n-r) + r + (m-r) = n + m = 6 + 5 = \mathbf{11}$ (constant!).

3.9 Problem Set 3.6 (Verbatim Questions and Solutions)

1. **(a)** If a 7 by 9 matrix has rank 5, what are the dimensions of the four subspaces? What is the sum of all four dimensions?
(b) If a 3 by 4 matrix has rank 3, what are its column space and left nullspace?

2. Find bases and dimensions for the four subspaces associated with A and B:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 8 \end{bmatrix}.$$

3. Find a basis for each of the four subspaces associated with A:

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

4. Construct a matrix with the required property or explain why this is impossible:
(a) Column space contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, row space contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.
(b) Column space has basis $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$, nullspace has basis $\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$.
(c) Dimension of nullspace $= 1 +$ dimension of left nullspace.
(d) Left nullspace contains $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, row space contains $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.
(e) Row space $=$ column space, nullspace $\neq$ left nullspace.

5. If $\mathbf{V}$ is the subspace spanned by $(1,1,1)$ and $(2,1,0)$, find a matrix A that has $\mathbf{V}$ as its row space. Find a matrix B that has $\mathbf{V}$ as its nullspace.

6. Without elimination, find dimensions and bases for the four subspaces for

$$A = \begin{bmatrix} 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}.$$

7. Suppose the 3 by 3 matrix A is invertible. Write down bases for the four subspaces for A, and also for the 3 by 6 matrix $B = \begin{bmatrix} A & A \end{bmatrix}$.

8. What are the dimensions of the four subspaces for A, B, and C, if I is the 3 by 3 identity matrix and 0 is the 3 by 2 zero matrix?

$$A = \begin{bmatrix} I & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} I & I \\ 0^T & 0^T \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 0 \end{bmatrix}.$$

9. Which subspaces are the same for these matrices of different sizes?

(a) $\begin{bmatrix} A \end{bmatrix}$ and $\begin{bmatrix} A \\ A \end{bmatrix}$ (b) $\begin{bmatrix} A \\ A \end{bmatrix}$ and $\begin{bmatrix} A & A \\ A & A \end{bmatrix}$. Prove that all three have the *same rank* r.

10. If the entries of a 3 by 3 matrix are chosen randomly between 0 and 1, what are the most likely dimensions of the four subspaces? What if the matrix is 3 by 5?

11. (*Important*) A is an m by n matrix of rank r. Suppose there are right sides $\boldsymbol{b}$ for which $A\boldsymbol{x} = \boldsymbol{b}$ has *no solution*.
(a) What are all inequalities ($<$ or $\leq$) that must be true between m, n, and r?
(b) How do you know that $A^T\boldsymbol{y} = \boldsymbol{0}$ has solutions other than $\boldsymbol{y} = \boldsymbol{0}$?

12. Construct a matrix with $(1,0,1)$ and $(1,2,0)$ as a basis for its row space and its column space. Why can't this be a basis for the row space and nullspace?

13. True or false (with a reason or a counterexample):
(a) If $m = n$ then the row space of A equals the column space.
(b) The matrices A and $-A$ share the same four subspaces.
(c) If A and B share the same four subspaces then A is a multiple of B.

14. Without computing A, find bases for its four fundamental subspaces:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 9 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

15. If you exchange the first two rows of A, which of the four subspaces stay the same? If $\boldsymbol{v} = (1,2,3,4)$ is in the left nullspace of A, write down a vector in the left nullspace of the new matrix.

16. Explain why $\boldsymbol{v} = (1,0,-1)$ *cannot be a row of A and also in the nullspace.*

END-OF-PAGE SOLUTION Problem Set 3.6 (Problems 1–16)

1. (a) $\dim C(A^T) = 5$, $\dim N(A) = 9-5 = 4$, $\dim C(A) = 5$, $\dim N(A^T) = 7-5 = 2$; sum $= 9+7 = \mathbf{16}$. **(b)** $C(A) = \mathbf{R}^3$, $N(A^T) = \{\boldsymbol{0}\}$.

2. For A ($r = 1$): $C(A^T)$ basis $\{(1,2,4)\}$ (dim 1); $N(A)$ basis $\{(-2,1,0),(-4,0,1)\}$ (dim 2); $C(A)$ basis $\{(1,2)\}$ (dim 1); $N(A^T)$ basis $\{(-2,1)\}$ (dim 1). For B ($r = 2$): $C(B^T)$ basis $\{(1,0,4),(0,1,0)\}$ (dim 2); $N(B)$ basis $\{(-4,0,1)\}$ (dim 1); $C(B)$ basis $\{(1,2),(2,5)\}$ (dim 2); $N(B^T) = \{\boldsymbol{0}\}$ (dim 0).

3. $r = 2$ (pivot cols 2, 4). $C(A^T)$: $\{(0,1,2,3,4),(0,0,0,1,2)\}$; $N(A)$: $\{(1,0,0,0,0),(0,-2,1,0,0),(0,-2,0,-2,1)\}$; $C(A)$: $\{(1,1,0),(3,4,1)\}$ (or cols 1,2 of L: $(1,1,0),(0,1,1)$); $N(A^T)$: $\{(1,-1,1)\}$.

4. (a) $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 2 & 5 \end{bmatrix}$. **(b) Impossible**: $n = 3$, $\dim C(A) = 1, \dim N(A) = 1$, $1+1 \neq 3$. **(c)** Any $m \times n$ matrix with $n - m = 1$ (e.g. 1×2). **(d)** $A = \begin{bmatrix} -3 \\ 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \end{bmatrix} = \begin{bmatrix} -9 & -3 \\ 3 & 1 \end{bmatrix}$. **(e) Impossible**: $C(A^T) = C(A) \implies N(A) = N(A^T)$ (orthogonal complements).

5. $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$; $B = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$.

6. For A ($r = 2$): $C(A^T)$ (dim 2): $\{(0,1,1,1),(0,1,0,1)\}$; $N(A)$ (dim 2): $\{(1,0,0,0),(0,-1,0,1)\}$; $C(A)$ (dim 2): $\{(3,0,1),(3,0,0)\}$; $N(A^T)$ (dim 1): $\{(0,1,0)\}$. For B ($r = 1$): $C(B^T)$ (dim 1): $\{(1)\}$; $N(B)$ (dim 0): $\emptyset$; $C(B)$ (dim 1): $\{(1,4,5)\}$; $N(B^T)$ (dim 2): $\{(-4,1,0),(-5,0,1)\}$.

7. For A: $C(A^T) = C(A) = \mathbf{R}^3$ (basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$), $N(A) = N(A^T) = \{\boldsymbol{0}\}$. For B: $C(B) = \mathbf{R}^3$,

$N(B^T) = \{\mathbf{0}\}$, $C(B^T)$ basis is rows of $\begin{bmatrix} I & I \end{bmatrix}$ (or $\begin{bmatrix} A & A \end{bmatrix}$), $N(B)$ basis is columns of $\begin{bmatrix} -I \\ I \end{bmatrix}$.

8. A $(3 \times 5, r = 3)$: $3, 2, 3, 0$. B $(5 \times 6, r = 3)$: $3, 3, 3, 2$. C $(3 \times 2, r = 0)$: $0, 2, 0, 3$.

9. (a) Same **row space** and **nullspace**. **(b)** Same **column space** and **left nullspace**. Thus $\text{rank}(A) = \text{rank} \begin{bmatrix} A \\ A \end{bmatrix} = \text{rank} \begin{bmatrix} A & A \\ A & A \end{bmatrix} = r$.

10. 3×3 $(r = 3)$: $3, 0, 3, 0$. 3×5 $(r = 3)$: $3, 2, 3, 0$.

11. (a) $r < m$ and $r \leq n$. **(b)** $\dim N(A^T) = m - r > 0$, so $N(A^T)$ has at least one free variable.

12. $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 4 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. A vector in both the row space and nullspace is perpendicular to itself ($\boldsymbol{v} \cdot \boldsymbol{v} = 0 \implies \boldsymbol{v} = \mathbf{0}$).

13. (a) False ($\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$). **(b) True**. **(c) False**: any two invertible 2×2 matrices share the same four subspaces ($\mathbf{R}^2, \{\mathbf{0}\}, \mathbf{R}^2, \{\mathbf{0}\}$).

14. $r = 3$: $C(A^T)$ basis is the 3 rows of U; $C(A)$ basis is the 3 columns of L; $N(A)$ basis is $\{(0, -1, 2, -1)\}$; $N(A^T) = \{\mathbf{0}\}$ (empty basis).

15. Row space and **nullspace** stay the same. New left-nullspace vector: $(\mathbf{2}, \mathbf{1}, \mathbf{3}, \mathbf{4})$.

16. If $\boldsymbol{v} = (1, 0, -1)$ is a row of A and $A\boldsymbol{v} = \mathbf{0}$, then the dot product of that row with $\boldsymbol{v}$ must be $\boldsymbol{v} \cdot \boldsymbol{v} = 1^2 + 0^2 + (-1)^2 = 2 = 0$, impossible!

17. Describe the four subspaces of $\mathbf{R}^3$ associated with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad I + A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

18. (Left nullspace) Add the extra column $\boldsymbol{b}$ and reduce A to echelon form:

$$\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 4 & 5 & 6 & b_2 \\ 7 & 8 & 9 & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & b_1 \\ 0 & -3 & -6 & b_2 - 4b_1 \\ 0 & 0 & 0 & b_3 - 2b_2 + b_1 \end{bmatrix}.$$

A combination of the rows of A has produced the zero row. What combination is it? Which vectors are in the nullspace of A^T and which are in the nullspace of A?

19. Following the method of Problem 18, reduce A to echelon form and look at zero rows. The $\boldsymbol{b}$ column tells which combinations you have taken of the rows:

$$\text{(a)} \begin{bmatrix} 1 & 2 & b_1 \\ 3 & 4 & b_2 \\ 4 & 6 & b_3 \end{bmatrix} \qquad \text{(b)} \begin{bmatrix} 1 & 2 & b_1 \\ 2 & 3 & b_2 \\ 2 & 4 & b_3 \\ 2 & 5 & b_4 \end{bmatrix}.$$

From the $\boldsymbol{b}$ column after elimination, read off $m - r$ basis vectors in the left nullspace.

20. **(a)** Check that the solutions to $A\boldsymbol{x} = \mathbf{0}$ are perpendicular to the rows:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} = ER.$$

(b) How many independent solutions to $A^T\boldsymbol{y} = \mathbf{0}$? Why is $\boldsymbol{y}^T$ the last row of E^{-1}?

21. Suppose A is the sum of two matrices of rank one: $A = \boldsymbol{u}\boldsymbol{v}^T + \boldsymbol{w}\boldsymbol{z}^T$.
(a) Which vectors span the column space of A? **(b)** Which vectors span the row space of A?
(c) The rank is less than 2 if _____ or if _____. **(d)** Compute A and its rank if $\boldsymbol{u} = \boldsymbol{z} = (1, 0, 0)$ and $\boldsymbol{v} = \boldsymbol{w} = (0, 0, 1)$.

22. Construct $A = \boldsymbol{u}\boldsymbol{v}^T + \boldsymbol{w}\boldsymbol{z}^T$ whose column space has basis $(1, 2, 4), (2, 2, 1)$ and whose row space has basis $(1, 0), (1, 1)$. Write A as (3 by 2) times (2 by 2).

23. Without multiplying matrices, find bases for the row and column spaces of A:

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 3 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}. \quad \text{How do you know from these shapes that } A \text{ cannot be invertible?}$$

24. (*Important*) $A^T\boldsymbol{y} = \boldsymbol{d}$ is solvable when $\boldsymbol{d}$ is in which of the four subspaces? The solution $\boldsymbol{y}$ is unique when the _____ contains only the zero vector.

25. True or false (with a reason or a counterexample):
(a) A and A^T have the same number of pivots. **(b)** A and A^T have the same left nullspace.
(c) If the row space equals the column space then $A^T = A$. **(d)** If $A^T = -A$ then the row space of A equals the column space.

26. (**Rank of** AB) If $AB = C$, the rows of C are combinations of the rows of _____. So the rank of C is not greater than the rank of _____. Since $B^TA^T = C^T$, the rank of C is also not greater than the rank of _____.

27. If a, b, c are given with $a \neq 0$, how would you choose d so that $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has rank 1? Find a basis for the row space and nullspace. Show they are perpendicular!

28. Find the ranks of the 8 by 8 checkerboard matrix B and the chess matrix C (p. 194). Find bases for the row space and left nullspace of B and C. Challenge problem: Find a basis for the nullspace of C.

29. Can tic-tac-toe be completed (5 ones and 4 zeros in A) so that $\text{rank}(A) = 2$ but neither side passed up a winning move?

30. **Challenge Problems:** If $A = \boldsymbol{u}\boldsymbol{v}^T$ is a 2 by 2 matrix of rank 1, redraw Figure 3.5 to show clearly the Four Fundamental Subspaces. If B produces those same four subspaces, what is the exact relation of B to A?

31. **M** is the space of 3 by 3 matrices. Multiply every matrix X in **M** by $A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$.

(a) Which matrices X lead to $AX =$ zero matrix? **(b)** Which matrices have the form AX for some matrix X? What are the dimensions of those two subspaces of **M**?

32. Suppose the m by n matrices A and B have the *same four subspaces*. If they are both in row reduced echelon form, prove that F must equal G: $A = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} I & G \\ 0 & 0 \end{bmatrix}$.

END-OF-PAGE SOLUTION Problem Set 3.6 (Problems 17–32)

17. For A: $C(A^T) = \text{span}\{(0,1,0),(0,0,1)\}$, $N(A) = \text{span}\{(1,0,0)\}$, $C(A) = \text{span}\{(1,0,0),(0,1,0)\}$, $N(A^T) = \text{span}\{(0,0,1)\}$. For $I + A$ (invertible): $C(I + A^T) = C(I + A) = \mathbf{R}^3$, $N(I + A) = N(I + A^T) = \{\mathbf{0}\}$.

18. $1(\text{row } 1) - 2(\text{row } 2) + 1(\text{row } 3) = \mathbf{0}^T$. Both $N(A^T)$ and $N(A)$ are all multiples of $(1, -2, 1)$.

19. **(a)** $b_3 - b_2 - b_1 = 0 \implies N(A^T)$ basis $\{(-1,-1,1)\}$. **(b)** $b_3 - 2b_1 = 0$ and $b_4 + b_2 - 3b_1 = 0 \implies N(A^T)$ basis $\{(-2,0,1,0),(-3,1,0,1)\}$.

20. **(a)** $\boldsymbol{s}_1 = (-1/2,1,0,0), \boldsymbol{s}_2 = (-1/4,0,-3,1)$ are orthogonal to $(4,2,0,1)$ and $(0,0,1,3)$. **(b)** $3 - 2 = \mathbf{1}$ independent solution; $E^{-1}A = R$ has row 3 equal to $\mathbf{0}^T$, so $(\text{row } 3 \text{ of } E^{-1})A = \mathbf{0}^T$.

21. **(a)** $\boldsymbol{u}$ and $\boldsymbol{w}$. **(b)** $\boldsymbol{v}$ and $\boldsymbol{z}$. **(c)** $\boldsymbol{u}, \boldsymbol{w}$ are dependent or $\boldsymbol{v}, \boldsymbol{z}$ are dependent. **(d)** $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $\text{rank}(A) = 2$.

22. $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 2 \\ 5 & 1 \end{bmatrix}$. **23.** $C(A^T)$ basis: $\{(3,0,3),(1,1,2)\}$; $C(A)$ basis: $\{(1,4,2),(2,5,7)\}$. A is 3×3 of rank $\leq 2 < 3$, so A is singular.

24. $\boldsymbol{d}$ in the **row space** $C(A^T)$; unique when the **left nullspace** $N(A^T)$ contains only **0**.

25. **(a) True**. **(b) False** (different sizes if $m \neq n$). **(c) False** ($\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ has $C(A) = C(A^T) = \mathbf{R}^2$). **(d) True**: $C(A^T) = C(-A) = C(A)$.

26. Rows of **B**; rank of **B**; rank of **A**. **27.** $d = \frac{bc}{a}$; row space basis $\{(a,b)\}$, nullspace basis $\{(-b,a)\}$, and $(a,b) \cdot (-b,a) = 0$.

28. $\text{rank}(B) = \mathbf{2}$ (row space basis: rows 1 and 2; $N(B^T)$ basis: 6 vectors $\boldsymbol{e}_k - \boldsymbol{e}_{k-2}$ for $k = 3, \ldots, 8$). $\text{rank}(C) = \mathbf{2}$ (row space basis: rows 1 and 2; $N(C^T)$ basis:

$e_3, e_4, e_5, e_6, e_7 - e_2, e_8 - e_1$). $N(C)$ has dimension $8 - 2 = 6$: 3 symmetric differences $e_1 - e_8, e_2 - e_7, e_3 - e_6$ plus 3 vectors orthogonal to $(1,1)$ and $(r,q), (n,q), (b,q)$ on cols $1, 2, 3, 4$.

29. Yes: $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ has 5 ones, 4 zeros, $\text{rank}(A) = 2$, and X (1s) wins on turn 9 without either player passing up an earlier winning move.

30. $B = cA$ for any nonzero scalar $c \neq 0$. **31. (a)** Each column of X is a multiple of $(1,1,1)$ ($\dim = 3$). **(b)** Each column of AX has components summing to 0 ($\dim = 6$). Dimensions add to $3 + 6 = 9 = \dim(\mathbf{M})$.

32. Same nullspace means $N(A) = N(B) \implies \begin{bmatrix} -F \\ I \end{bmatrix}$ and $\begin{bmatrix} -G \\ I \end{bmatrix}$ span the same space. Multiply by any vector $\boldsymbol{c}$: since the bottom $n - r$ components are $I\boldsymbol{c} = \boldsymbol{c}$, the top r components must also match: $-F\boldsymbol{c} = -G\boldsymbol{c}$ for all $\boldsymbol{c} \implies F = G$.

Module 3: Unique Solved Problem Archetypes & Formula Sheet

1. Complete Solution to $Ax = b$ (Section 3.4)

- **Augmented Row Reduction:** $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix} \to \begin{bmatrix} R & \boldsymbol{d} \end{bmatrix}$. Solvable iff $d_{r+1} = \cdots = d_m = 0$ ($\boldsymbol{y}^T\boldsymbol{b} = 0\ \forall \boldsymbol{y} \in N(A^T)$).
- **Complete Solution:** $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$, where $(\boldsymbol{x}_p)_{\text{free}} = 0$, $(\boldsymbol{x}_p)_{\text{pivot}} = \boldsymbol{d}_{1:r}$, and $\boldsymbol{x}_n = c_1\boldsymbol{s}_1 + \cdots + c_{n-r}\boldsymbol{s}_{n-r}$.

2. Fundamental Theorem of Linear Algebra, Part 1 (Sections 3.5–3.6)

- **Subspace Dimensions:** $\dim C(A^T) = \dim C(A) = r$, $\dim N(A) = n - r$, $\dim N(A^T) = m - r$.
- **Factored Matrix Shortcut ($A = LU = E^{-1}R$, Ex 3.6 A, PS 3.6 #3, #14, #20):** Row space basis = first r rows of R; Column space basis = first r columns of E^{-1} (or pivot columns of A); Nullspace basis = $n - r$ special solutions of $R\boldsymbol{x} = \boldsymbol{0}$; Left nullspace basis = last $m - r$ rows of E.